

GRADIENT DESCENT FOR CONVEX AND SMOOTH NOISY OPTIMIZATION

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ABSTRACT. We study gradient descent with backtracking line search (GD-BLS) to solve the noisy optimization problem $\theta_\star := \operatorname{argmin}_{\theta \in \mathbb{R}^d} \mathbb{E}[f(\theta, Z)]$, imposing that the objective function $F(\theta) := \mathbb{E}[f(\theta, Z)]$ is strictly convex but not necessarily L -smooth. Assuming that $\mathbb{E}[\|\nabla_\theta f(\theta_\star, Z)\|^2] < \infty$, we first prove that sample average approximation based on GD-BLS allows to estimate $\theta_\star$ with an error of size $\mathcal{O}_{\mathbb{P}}(B^{-0.25})$, where B is the available computational budget. We then show that we can improve upon this rate by stopping the optimization process earlier when the gradient of the objective function is sufficiently close to zero, and use the residual computational budget to optimize, again with GD-BLS, a finer approximation of F . By iteratively applying this strategy J times we establish that we can estimate $\theta_\star$ with an error of size $\mathcal{O}_{\mathbb{P}}(B^{-\frac{1}{2}(1-\delta^J)})$, where $\delta \in (1/2, 1)$ is a user-specified parameter. More generally, we show that if $\mathbb{E}[\|\nabla_\theta f(\theta_\star, Z)\|^{1+\alpha}] < \infty$ for some known $\alpha \in (0, 1]$ then this approach, which can be seen as a retrospective approximation algorithm with a fixed computational budget, allows to learn $\theta_\star$ with an error of size $\mathcal{O}_{\mathbb{P}}(B^{-\frac{\alpha}{1+\alpha}(1-\delta^J)})$, where $\delta \in (2\alpha/(1+3\alpha), 1)$ is a tuning parameter. Beyond knowing α , achieving the aforementioned convergence rates does not require to tune the algorithms' parameters according to the specific functions F and f at hand, and we exhibit a simple noisy optimization problem for which stochastic gradient is not guaranteed to converge while the algorithms discussed in this work are.

1. INTRODUCTION

1.1. Set-up and problem formulation. We consider the problem of computing

$$(1) \quad \theta_\star := \operatorname{argmin}_{\theta \in \mathbb{R}^d} F(\theta), \quad F(\theta) = \mathbb{E}[f(\theta, Z)],$$

where Z is a random variable taking its values in a Polish space $\mathbf{Z}$ and where $f : \mathbb{R}^d \times \mathbf{Z} \rightarrow \mathbb{R}$ is a measurable function such that $\nabla_\theta f(\theta, z)$ exists for all $(\theta, z) \in \Theta \times \mathbf{Z}$. We assume that only the function f can be evaluated point-wise and that to solve (1) we have at our disposal a sequence $(Z_i)_{i \geq 1}$ of i.i.d. copies of Z . All the random variables are assumed to be defined on a complete probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and the function F is assumed to be strictly convex on $\mathbb{R}^d$, and thus the optimization problem (1) is well-defined. Formal assumptions on F are given in Section 2.2, and we stress that in this work the function $\theta \mapsto f(\theta, Z)$ is not supposed to be $\mathbb{P}$ -almost surely convex.

Due to their ease of implementation and low computational cost, stochastic gradient (SG) algorithms arguably constitute the main set of tools used in practice for solving this type of optimization problems [16]. Stochastic gradient methods

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are for instance widely applied for training machine learning algorithms [3] or for performing statistical inference in large datasets [18]. In order to guarantee that a SG algorithm is stable when deployed on an unbounded search space it is commonly assumed that the variance of the norm of the stochastic gradient $\nabla_{\theta}f(\theta, Z)$ is bounded and that the function F is L -smooth, that is that the gradient of F is L -Lipschitz (see for instance [15] and [8] for a review of convergence results for SG). Under the additional assumption that F is twice differentiable, the condition that F is L -smooth can be replaced by the weaker assumption that F is (L_0, L_1) -smooth, in the sense that $\|\nabla^2 F(\theta)\| \leq L_0 + L_1\|\nabla F(\theta)\|$ for all $\theta \in \mathbb{R}^d$ and some finite constants L_0 and L_1 (see Assumption 3 in [21]). The assumption that the variance of $\|\nabla_{\theta}f(\theta, Z)\|$ is bounded can also be weakened. Notably, letting C be a finite constant, it can instead be assumed that, for all $\theta \in \mathbb{R}^d$, this variance is bounded by $C + C\|\nabla F(\theta)\|^2$ (see e.g. Assumption 4.3 in [3] and Assumption 2 in [6]) or that $\mathbb{E}[\|\nabla_{\theta}f(\theta, Z)\|^2] \leq C + C\|\nabla F(\theta)\|^2$ (see e.g. Assumption 2 in [9]). Convergence results for SG that hold under the weaker condition that $\mathbb{E}[\|\nabla_{\theta}f(\theta, Z)\|^{1+\alpha}] < \infty$ for some $\alpha \in (0, 1]$ have been obtained by some authors, see for instance [16, 20].

These assumptions used to establish the convergence of SG algorithms may however be violated even for simple optimization problems. For instance, as shown in [16], both the assumption that F is L -smooth and the assumption that the variance of $\|\nabla_{\theta}f(\theta, Z)\|$ is bounded, as well as their variants mentioned in the previous paragraph, may not hold for parameter inference in Poisson regression models. More precisely, Proposition 3 in [16] shows the following result:

Proposition 1. *Consider the $d = 1$ dimensional optimization problem (1) where $Z = (X, Y)$, with X and Y two independent Poisson random variables such that $\mathbb{E}[X] = \mathbb{E}[Y] = 1$, and where the function $f : \mathbb{R}^d \times \mathcal{Z} \rightarrow \mathbb{R}$ is defined by*

$$f(\theta, z) = -z_2 z_1 \theta + \exp(\theta z_1), \quad \theta \in \mathbb{R}, \quad z \in \mathcal{Z} := \mathbb{R}^2.$$

Then, the function F is strictly convex, twice continuously differentiable and we have $\text{Var}(\nabla_{\theta}f(\theta, Z)) < \infty$ for all $\theta \in \mathbb{R}$. However, F is neither globally L -smooth nor (L_0, L_1) -smooth, and there exists no finite constant C such that we have $\text{Var}(\nabla_{\theta}f(\theta, Z)) \leq C + C\|\nabla F(\theta)\|^2$ for all $\theta \in \mathbb{R}$ and there exists no finite constant C such that we have $\mathbb{E}[\|\nabla_{\theta}f(\theta, Z)\|^2] \leq C + C\|\nabla F(\theta)\|^2$ for all $\theta \in \mathbb{R}$ (and thus $\sup_{\theta \in \mathbb{R}} \text{Var}(\nabla_{\theta}f(\theta, Z)) = \infty$).

To the best of our knowledge, [16] provides the only existing general convergence result for SG on unbounded spaces that holds under the assumption that F is only locally L -smooth, and which allows the variance of $\|\nabla_{\theta}f(\theta, Z)\|$ to be unbounded. [16] actually only assumes that, for some $\alpha \in (0, 1]$, the gradient of F is locally α -Hölder continuous and that $\mathbb{E}[\|\nabla_{\theta}f(\theta, Z)\|^{1+\alpha}]$ is finite for all $\theta \in \mathbb{R}^d$. However, to guarantee the stability of SG, the result in this reference requires that the Hölder constant of ∇F on a ball centred at θ and of size $\mathbb{E}[\|\nabla_{\theta}f(\theta, Z)\|^{1+\alpha}]^{\frac{1}{1+\alpha}}$ does not increase too quickly as $\|\theta\| \rightarrow \infty$ (see Assumption 5 in [16]). This condition appears to be quite strong since, as we show in Proposition 2, it is not satisfied for the optimization problem considered in Proposition 1.

Proposition 2. *Consider the set-up of Proposition 1. Then, there exists no $\alpha \in (0, 1]$ such that Assumption 5 used in [16] to establish the convergence of SG holds.*

Finally, we note that [7] studies SG algorithms under the assumption that F is locally L -smooth and allowing $\mathbb{E}[\|\nabla_{\theta}f(\theta, Z)\|^2]$ to be unbounded but, informally

speaking, this reference only shows that SG will converge to θ_* if the starting value of the algorithm is sufficiently close to that point.

While we previously focused on SG methods, it is worth mentioning that stochastic proximal point methods (see [2] and references therein) form another popular set of tools for solving optimization problems as defined in (1). These algorithms have the advantage of having stronger theoretical properties than SG approaches but, on the other hand, they usually have the drawback to require, at each iteration, to use a numerical method to approximate the solution of an optimization problem. The Poisson regression example introduced in Proposition 1 provides a simple illustration of this claim: convergence results for the stochastic proximal point method applied to this problem have been obtained by [2] but, for this example, there is no explicit solution to the optimization problem that needs to be solved at each iteration of the algorithm. We finally note that the results derived in [2] in fact apply to a broader class of algorithms, called model-based stochastic proximal point methods. In these algorithms, the choice of the model determines the optimization problem that needs to be solved at each iteration, and [2] uses the Poisson regression example as a motivating example for what they call the truncated model. However, the authors do not show that the resulting algorithm is stable, that is that their key condition (C.iv) used to establish the stability of model-based stochastic proximal point methods is satisfied.

Consequently, as illustrated by the Poisson regression example of Proposition 1, there are seemingly simple noisy optimization problems on $\mathbb{R}^d$, where the objective function F is strictly convex and $\mathbb{E}[\|\nabla_\theta f(\theta, Z)\|^2]$ is finite for all $\theta \in \mathbb{R}^d$, which cannot be solved (easily) with theoretical guarantees using existing methods.

1.2. Contribution of the paper. In this paper we first show that gradient descent with backtracking line search (GD-BLS) can be used to solve (1) even in situations where F is not L -smooth and where $\sup_{\theta \in \mathbb{R}^d} \mathbb{E}[\|\nabla_\theta f(\theta, Z)\|^{1+\alpha}] = \infty$ for all $\alpha \in (0, 1]$.

More precisely, we assume throughout this work that there is a computational cost $C_{\text{grad}} \in \mathbb{N}$ for computing $\nabla_\theta f(\theta, z)$ and a computational cost $C_{\text{eval}} \in \mathbb{N}$ for computing $f(\theta, z)$ for a pair $(\theta, z) \in \mathbb{R}^d \times \mathcal{Z}$. Then, we first show that, given the available computational budget $B \in \mathbb{N}$, by applying GD-BLS to minimize the random function $F_{n_1(B)} := \frac{1}{n_1(B)} \sum_{i=1}^{n_1(B)} f(\cdot, Z_i)$ we obtain a consistent estimator $\hat{\theta}_{1,B}$ of θ_* . The convergence rate of this approach, which is a particular type of sample average approximation algorithms (see e.g. [10]), depends on the choice of the function n_1 and, assuming for now that $\mathbb{E}[\|\nabla_\theta f(\theta_*, Z)\|^2] < \infty$, the upper bound we obtain for this rate is minimized when $n_1(B) \approx B^{1/2}$, in which case $\|\hat{\theta}_{1,B} - \theta_*\| = \mathcal{O}_{\mathbb{P}}(B^{-0.25})$. Remark that the $B^{-0.25}$ convergence rate obtained for $\hat{\theta}_{1,B}$ is slower than the standard $B^{-0.5}$ convergence rate (see Proposition 4).

In order to save computational resources gradient descent algorithms are often stopped if a pre-specified convergence criterion is reached, typically when the gradient of the objective function is sufficiently close to zero. Following this approach, we propose to allow the optimization of $F_{n_1(B)}$ to stop earlier (i.e. before the computational budget B is exhausted) and, if any, to use the remaining computational resources to improve the estimation of θ_* . In this work the optimization of $F_{n_1(B)}$ is stopped if its gradient falls below some threshold $\tau_1(B)$ and the residual computational budget is used to apply GD-BLS on a finer approximation $F_{n_2(B)}$ of F

(i.e. $n_2(B) > n_1(B)$), using the estimate of θ_* obtained by minimizing $F_{n_1(B)}$ as initial value. With a careful choice for the functions n_1 , τ_1 and n_2 we show that the resulting estimate $\hat{\theta}_{2,B}$ of θ_* has an error of size $B^{-0.5(1-\delta^2)}$, with $\delta \in (1/2, 1)$ a user-specified parameter.

The approach outlined in the previous paragraph can be generalized as follows to any $J \geq 2$ applications of GD-BLS on increasingly finer approximations of F : For $j = 1, \dots, J-1$ we stop earlier the optimization of the function $F_{n_j(B)}$ by GD-BLS if its gradient becomes smaller than some threshold $\tau_j(B)$ and, if any, the remaining computational resources are used to repeat this process to the function $F_{n_{j+1}(B)}$, using as starting value the estimate $\tilde{\theta}_{j,B}$ of θ_* obtained by minimizing $F_{n_j(B)}$. For a suitable choice of functions $(n_j)_{j \geq 1}$ and $(\tau_j)_{j \geq 1}$, and a user-specified parameter $\delta \in (1/2, 1)$, we prove that the estimator $\hat{\theta}_{J,B}$ of θ_* defined by this approach has an error of size $\mathcal{O}_{\mathbb{P}}(B^{-\frac{1}{2}(1-\delta^J)})$ while, for all $j \in \{1, \dots, J-1\}$, the estimator $\tilde{\theta}_{j,B}$ converges to θ_* at rate $B^{-\frac{1}{2}(1-\delta^j)}$. The assumptions under which these results are obtained are shown to be satisfied for the optimization problem considered in Proposition 1, and to achieve the aforementioned convergence rates it is not needed to tune the parameters of the algorithms according to the specific functions F and f at hand.

In practice, to observe the $B^{-\frac{1}{2}(1-\delta^J)}$ convergence rate of $\hat{\theta}_{J,B}$ it is necessary that the optimization of $F_{n_j(B)}$ stops earlier for all $j \in \{1, \dots, J-1\}$, and thus that GD-BLS is applied to the function $F_{n_J(B)}$. Informally speaking, if we denote by J_B the smallest value of $j \in \{1, \dots, J\}$ such that $\tilde{\theta}_{j,B} = \hat{\theta}_{J,B}$, we therefore expect the observed convergence rate for $\hat{\theta}_{J,B}$ to be $B^{-\frac{1}{2}(1-\delta^{J_B})}$. In this work we do not study theoretically the behaviour of J_B but the numerical experiments suggest that, as B increases, this random variable ‘grows’ at speed $\log(B)$.

In some applications the variance of the noisy gradient is not finite, and in this work the procedure described above is studied under the weaker assumption that $\mathbb{E}[\|\nabla_{\theta} f(\theta_*, Z)\|^{1+\alpha}] < \infty$ for some $\alpha \in (0, 1]$. In this scenario, the user needs to choose a noise parameter $\alpha' \in (0, 1]$ and a $\delta \in (2\alpha'/(1+3\alpha'), 1)$, and the proposed approach for computing θ_* is proved to be consistent for any $\alpha' \in (0, 1]$. In addition, if $\alpha' = \alpha$ then its convergence rate for computing θ_* is shown to be $B^{-\frac{\alpha}{1+\alpha}(1-\delta^J)}$. By contrast, in this context and as shown in Proposition 4, under the assumptions imposed on (1) in the next section $B^{-\frac{\alpha}{1+\alpha}}$ is the best possible convergence rate for estimating θ_* .

1.3. Connection with retrospective approximation. The proposed J -step procedure can be seen as a retrospective approximation (RA) algorithm with a fixed computational budget and using GD-BLS as deterministic optimizer (see [10] for a survey on RA methods). Indeed, given a sequence $(n'_j)_{j \geq 1}$ in $\mathbb{N}$ and a sequence $(\tau'_j)_{j \geq 1}$ in $(0, \infty)$, RA based on GD-BLS amounts to applying the above J -step procedure with $n_j(B) = n'_j$ and $\tau_j(B) = \tau'_j$ for all $B \in \mathbb{N}$ and $j \in \mathbb{N}$, and with $B = \infty$ (so that the optimization of $F_{n'_j}$ is performed for all $j \in \{1, \dots, J\}$). Unlike in the proposed approach, the computational budget B_J consumed by an RA algorithm is therefore random. The behaviour of $\mathbb{E}[B_J]$ and the convergence rate, as $J \rightarrow \infty$, of the resulting estimator θ_J of θ_* have been recently studied by [13] under general conditions on the optimization routine $\mathcal{R}$ used to optimize $F_{n'_j}$.

for all $j \in \{1, \dots, J\}$. However, the results presented in this reference are obtained under conditions which seem hard to fulfil if F is not (globally) L -smooth.

1.4. Outline of the paper. The rest of the paper is organized as follows. In Section 2 we state the assumptions we impose on (1) and present some preliminary results. The use of GD-BLS for solving noisy optimization problems is studied in Section 3. In Section 4 we illustrate the convergence results derived in this work with some numerical experiments and Section 5 concludes. All the proofs are gathered in Appendix A.

2. NOTATION, ASSUMPTIONS AND SOME PRELIMINARY RESULTS

2.1. Notation. To simplify the notation, for all $(\theta, z) \in \mathbb{R}^d \times \mathbf{Z}$ we use the shorthand $\nabla^k f(\theta, z)$ for $\nabla_{\theta}^k f(\theta, z)$ for all $k \in \{0, 1, 2\}$, with the convention that for $k = 0$ we have $\nabla^k f(\theta, z) = f(\theta, z)$. For all $n \in \mathbb{N}$ and compact set $\Theta \subset \mathbb{R}^d$ we let $\hat{\theta}_{\Theta, n} \in \operatorname{argmin}_{\theta \in \Theta} F_n(\theta)$ and we denote by $\lambda_{\min}(M)$ and $\lambda_{\max}(M)$ the smallest and largest eigenvalues of a $d \times d$ matrix M , respectively. To introduce some further notation let $(X_t)_{t \geq 1}$ be a sequence of $\mathbb{R}$ -valued random variables. Then, we write $X_t = o_{\mathbb{P}}(1)$ if and only if $X_t \rightarrow 0$ in $\mathbb{P}$ -probability, that is if and only if $\lim_{t \rightarrow \infty} \mathbb{P}(|X_t| \geq \epsilon) = 0$ for all $\epsilon > 0$, and for a sequence $(a_t)_{t \geq 1}$ in $(0, \infty)$ we write $X_t = \mathcal{O}_{\mathbb{P}}(a_t)$ if and only if for all $\epsilon \in (0, 1)$ there exists a finite constant M_{ϵ} such that $\limsup_{t \rightarrow \infty} \mathbb{P}(|X_t/a_t| \geq M_{\epsilon}) \leq \epsilon$.

2.2. Assumptions. We consider Assumptions A1 and A2 on the optimization problem (1):

Assumption A1. The function F is locally strongly convex, that is, for every compact set $\Theta \subset \mathbb{R}^d$ there exists a constant $c_{\Theta} > 0$ such that, for all $(\theta_1, \theta_2) \in \Theta^2$,

$$F(a\theta_1 + (1-a)\theta_2) \leq aF(\theta_1) + (1-a)F(\theta_2) - \frac{c_{\Theta}}{2}a(1-a)\|\theta_1 - \theta_2\|^2, \quad \forall a \in [0, 1].$$

Remark that under A1 we have $F(a\theta_1 + (1-a)\theta_2) < aF(\theta_1) + (1-a)F(\theta_2)$ for all $a \in [0, 1]$ and all $(\theta_1, \theta_2) \in \mathbb{R}^d \times \mathbb{R}^d$, meaning that the function F is strictly convex on $\mathbb{R}^d$. Consequently, under A1, the optimization problem (1) is well-defined.

Assumption A2. The function $\theta \mapsto f(\theta, z)$ is twice continuously differentiable on $\mathbb{R}^d$ for all $z \in \mathbf{Z}$, and

$$\mathbb{E}[\|\nabla^2 f(\theta_*, Z)\|] < \infty, \quad \mathbb{E}[\|\nabla f(\theta_*, Z)\|^{1+\alpha}] < \infty$$

for some $\alpha \in (0, 1]$. Moreover, for all compact set $\Theta \subset \mathbb{R}^d$ there exists a measurable function $M_{\Theta} : \mathbf{Z} \mapsto [0, \infty)$ such that $\mathbb{E}[M_{\Theta}(Z)] < \infty$ and such that

$$\|\nabla^2 f(\theta, Z) - \nabla^2 f(\theta', Z)\| \leq M_{\Theta}(Z)\|\theta - \theta'\|, \quad \forall (\theta, \theta') \in \Theta^2, \quad \mathbb{P} - a.s.$$

Assumptions A1–A2 are quite weak. In particular, as shown in Proposition 3, they are satisfied for the optimization problem considered in Propositions 1–2.

Proposition 3. *Consider the set-up of Proposition 1. Then, Assumptions A1–A2 are satisfied with $\alpha = 1$.*

2.3. Some implications of Assumptions A1–A2. Lemma 1 notably implies that, under A2, the random function $\theta \mapsto f(\theta, Z)$ is $\mathbb{P}$ -a.s. locally L -smooth.

Lemma 1. *Assume that A2 holds. Then, for all compact set $\Theta \subset \mathbb{R}^d$ there exists a measurable function $M'_\Theta : Z \mapsto [0, \infty)$ such that $\mathbb{E}[M'_\Theta(Z)] < \infty$ and such that, for all $k \in \{0, 1, 2\}$, we have*

$$\|\nabla^k f(\theta, Z) - \nabla^k f(\theta', Z)\| \leq M'_\Theta(Z) \|\theta - \theta'\|, \quad \forall (\theta, \theta') \in \Theta^2, \quad \mathbb{P} - a.s.$$

Remark that, in the set-up of Lemma 1, for any $\epsilon > 0$ it is possible that $\mathbb{E}[M'_\Theta(Z)^{1+\epsilon}] = \infty$. Therefore, under A2, the quantity $\mathbb{E}[\|\nabla f(\theta, Z)\|^{1+\alpha}]$ is assumed to be finite for some $\alpha \in (0, 1]$ only if $\theta = \theta_*$. In addition, it is worth noting that no conditions are imposed on the behaviour of the function M'_Θ as the size of the set Θ increases.

Lemma 2 shows that, under A2, we can swap the expectation and the differential operators.

Lemma 2. *Assume that A2 holds. Then, the function F is twice continuously differentiable on $\mathbb{R}^d$ and*

$$\nabla^k F(\theta) = \mathbb{E}[\nabla^k f(\theta, Z)], \quad \forall \theta \in \mathbb{R}^d, \quad \forall k \in \{0, 1, 2\}.$$

By combining Lemma 1 and Lemma 2 it follows that under A2 the function F is locally L -smooth. More precisely, under A2, for any compact set $\Theta \subset \mathbb{R}^d$ we have

$$\|\nabla F(\theta) - \nabla F(\theta')\| \leq \mathbb{E}[M'_\Theta(Z)] \|\theta - \theta'\|, \quad \forall (\theta, \theta') \in \Theta^2$$

with the function M'_Θ as in Lemma 1. We stress that, unlike in [16], no conditions are imposed on the behaviour of the Lipschitz constant $\mathbb{E}[M'_\Theta(Z)]$ as the size of the set Θ increases.

Lemma 3 shows that the random function F_n converges uniformly to F on any compact subset of $\mathbb{R}^d$.

Lemma 3. *Assume that A2 holds. Then, for all compact set $\Theta \subset \mathbb{R}^d$ we have*

$$\lim_{n \rightarrow \infty} \sup_{\theta \in \Theta} \|\nabla^k F_n(\theta) - \nabla^k F(\theta)\| = 0, \quad \mathbb{P} - a.s. \quad \forall k \in \{0, 1, 2\}.$$

Lemma 4 provides a useful result on the asymptotic behaviour of the largest and smallest eigenvalues of the random matrix $\nabla^2 F_n(\theta)$.

Lemma 4. *Assume that A2 holds. Then, for all compact set $\Theta \subset \mathbb{R}^d$ there exists a $\theta_\Theta \in \Theta$ such that*

$$\liminf_{n \rightarrow \infty} \left(\inf_{\theta \in \Theta} \lambda_{\min}(\nabla^2 F_n(\theta)) \right) \geq \lambda_{\min}(\mathbb{E}[\nabla^2 f(\theta_\Theta, Z)]), \quad \mathbb{P} - a.s.$$

and a $\theta'_\Theta \in \Theta$ such that

$$\limsup_{n \rightarrow \infty} \left(\sup_{\theta \in \Theta} \lambda_{\max}(\nabla^2 F_n(\theta)) \right) \leq \lambda_{\max}(\mathbb{E}[\nabla^2 f(\theta'_\Theta, Z)]), \quad \mathbb{P} - a.s.$$

Under A1 and by combining Lemma 2 and Lemma 4, we readily obtain the following result which notably shows that if $\Theta \subset \mathbb{R}^d$ is a compact and convex set then the random function F_n is strictly convex on Θ with $\mathbb{P}$ -probability tending to one.

Corollary 1. Assume that A1–A2 hold. Then, for all compact set $\Theta \subset \mathbb{R}^d$ there exists a constant $c_\Theta \in (0, 1)$ such that, $\mathbb{P}$ -a.s.,

$$\liminf_{n \rightarrow \infty} \left(\inf_{\theta \in \Theta} \lambda_{\min}(\nabla^2 F_n(\theta)) \right) \geq c_\Theta, \quad \limsup_{n \rightarrow \infty} \left(\sup_{\theta \in \Theta} \lambda_{\max}(\nabla^2 F_n(\theta)) \right) \leq \frac{1}{c_\Theta}.$$

Lemma 5 establishes that, under A1 and A2, by using the first n elements of $(Z_i)_{i \geq 1}$ we can learn θ_* at rate $n^{-\frac{\alpha}{1+\alpha}}$, with $\alpha \in (0, 1]$ as in A2. In addition to be used to establish our main results, this lemma will allow us to discuss the convergence rate of the proposed estimators.

Lemma 5. Assume that A1–A2 hold and let $\Theta \subset \mathbb{R}^d$ be a compact and convex set such that $\theta_* \in \dot{\Theta}$. Then, $\|\theta_* - \hat{\theta}_{\Theta, n}\| = \mathcal{O}_{\mathbb{P}}(n^{-\frac{\alpha}{1+\alpha}})$, with $\alpha \in (0, 1]$ as in A2.

Finally, Proposition 4 shows that, under A1–A2 and in the context of this work (i.e. noisy optimization under fixed computations budget), the best possible convergence rate for an estimator of θ_* is $B^{-\frac{\alpha}{1+\alpha}}$, with $\alpha \in (0, 1]$ as in A2.

Proposition 4. Let $\alpha' \in (0, 1]$. Then, for any sequence $(\hat{\theta}_B)_{B \geq 1}$ of estimators of θ_* such that, for all $B \in \mathbb{N}$, $\hat{\theta}_B$ depends on $(Z_i)_{i \geq 1}$ only through $k_B \in \mathbb{N}_0$ evaluations of $f(\cdot, \cdot)$ and $k'_B \in \mathbb{N}_0$ evaluations of $\nabla f(\cdot, \cdot)$, where $k_B C_{\text{eval}} + k'_B C_{\text{grad}} \leq B$, there is a noisy optimization problem (1) such that A1–A2 hold with $\alpha = \alpha'$ and such that, for some constants $\delta \in (0, 1)$ and $C \in (0, \infty)$, we have

$$\limsup_{B \rightarrow \infty} \mathbb{P} \left(\|\hat{\theta}_B - \theta_*\| > CB^{-\frac{\alpha}{1+\alpha}} \right) \geq \delta.$$

2.4. Gradient descent with backtracking line search. Algorithm 1 provides a generic description of a gradient descent algorithm that can be used to minimize a differentiable function $g : \mathbb{R}^d \rightarrow \mathbb{R}$. The convergence behaviour of Algorithm 1 depends on the value of the step-size v_t chosen on its Line 2, and our theoretical analysis of gradient descent applied to noisy optimization problems builds on the following result.

Proposition 5. Let $g : \mathbb{R}^d \rightarrow \mathbb{R}$ be a differentiable function, $x_0 \in \mathbb{R}^d$, and for all $T \in \mathbb{N}$ let x_T be as computed by Algorithm 1 with $(v_t)_{t \geq 1}$ such that, for some constant $c > 0$,

$$(2) \quad g(x_t) \leq g(x_{t-1}) - \frac{v_t}{2} \|\nabla g(x_{t-1})\|^2, \quad v_t \geq c, \quad \forall t \geq 1.$$

Assume that there exists a convex set $K \subseteq \mathbb{R}^d$ such that (i) $x_t \in K$ for all $t \geq 0$, (ii) the function g is strictly convex on K and (iii) there exists an $x_* \in K$ such that $g(x_*) = \inf_{x \in K} g(x)$. Then,

$$g(x_T) - g(x_*) \leq \frac{\|x_0 - x_*\|^2}{2cT}, \quad \forall T \geq 1.$$

If the function g is L -smooth (assuming without loss of generality that $L > 1$) then (2) holds for instance when $v_t = v$ for all $t \geq 1$ and some $v \in (0, 1/L]$ (see e.g. the proof of Theorem 2.1.14 in [12, page 80]). In practice, the value of L is however rarely known and backtracking line search, described in Algorithm 2, is a simple procedure that can be used to automatically compute, at each iteration t , a value for v_t which ensures that (2) holds with $c = \beta/L$, where $\beta \in (0, 1)$ is a parameter of Algorithm 2. This property of backtracking line search is formalized in Proposition 6, which plays a key role in the proofs of our main results.

Algorithm 1 Generic gradient descent algorithm**Input:** Starting value $x_0 \in \mathbb{R}^d$ and number of iterations $T \in \mathbb{N}$.

```

1: for  $t = 1, \dots, T$  do
2:   Choose a set-size  $v_t$ 
3:    $x_t = x_{t-1} - v_t \nabla g(x_{t-1})$ 
4: end for
Return:  $x_T$ 

```

Algorithm 2 Backtracking line search**Input:** $x_{t-1} \in \mathbb{R}^d$ and backtracking parameter $\beta \in (0, 1)$.

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1:  $g_{t-1} \leftarrow \nabla g(x_{t-1})$ 
2:  $v_t = 1$ 
3: while  $g(x_{t-1} - v_t g_{t-1}) > g(x_{t-1}) - \frac{v_t}{2} \|g_{t-1}\|^2$  do
4:    $v_t \leftarrow \beta v_t$ 
5: end while
6: return:  $v_t$ 

```

Proposition 6. Let $g : \mathbb{R}^d \rightarrow \mathbb{R}$ be a differentiable function, $x_0 \in \mathbb{R}^d$, $\beta \in (0, 1)$, and for all $T \in \mathbb{N}$ let x_T be as computed by Algorithm 1 where Algorithm 2 is used to compute v_t for all $t \geq 1$. In addition, let $\tilde{K} \subseteq \mathbb{R}^d$ be a convex set containing the set $\{x \in \mathbb{R}^d : g(x) \leq g(x_0)\}$ and let

$$K = \left\{ x \in \mathbb{R}^d : \exists x' \in \tilde{K} \text{ such that } \|x - x'\| \leq \sup_{\tilde{x} \in \tilde{K}} \|\nabla g(\tilde{x})\| \right\}.$$

Assume that the function g is strictly convex on K and that there exists a constant $L_K \in [1, \infty)$ such that $\|\nabla g(x) - \nabla g(x')\| \leq L_K \|x - x'\|$ for all $(x, x') \in K^2$. Then, (2) holds with $c = \beta/L_K$.

Given the large literature on convex optimization, the conclusions of Propositions 5–6 are probably known. We however failed to find a reference establishing similar results and therefore the two propositions are proved in Appendix A.

3. USING GRADIENT DESCENT FOR NOISY OPTIMIZATION

3.1. A simple application of gradient descent to noisy optimization. Recall that it is assumed in this work that there is a computational cost $C_{\text{grad}} \in \mathbb{N}$ for computing $\nabla f(\theta, z)$ and a computational cost $C_{\text{eval}} \in \mathbb{N}$ for computing $f(\theta, z)$ for a given pair $(\theta, z) \in \mathbb{R}^d \times \mathcal{Z}$. Then, for a given computational budget $B \in \mathbb{N}$, a simple way to estimate θ_* is to apply Algorithm 1 to minimize the function $g = F_{n(B)}$ for some $n(B) \in \mathbb{N}$, using at each iteration the backtracking procedure described in Algorithm 2 for choosing the step-size and running the algorithm until the computational resources are exhausted. Algorithm 3 (with $\tau = 0$) provides a precise description of this procedure.

In order for the resulting estimator $\tilde{\theta}_B$ of θ_* to be consistent as $B \rightarrow \infty$, it is clear that we must have (i) $n(B) \rightarrow \infty$, so that $F_{n(B)} \rightarrow F$, and (ii) $n(B)/B \rightarrow 0$, so that the number of iterations of the algorithm converges to infinity. Theorem 1 formalizes this intuitive reasoning and provides an upper bound on the convergence rate of $\|\tilde{\theta}_B - \theta_*\|$.

Algorithm 3

Input: Starting value $\theta_0 \in \mathbb{R}^d$, sample size $n \in \mathbb{N}_0$, computational budget $B \in \mathbb{N}_0$, tolerance parameter $\tau \in [0, \infty)$ and backtracking parameter $\beta \in (0, 1)$.

```

1: Let  $t = 0$ ,  $\theta_{B,t} = \theta_0$  and  $\tilde{B} = B$ 
2: if  $\tilde{B} \geq nC_{\text{grad}}$  then
3:   Let  $G_t = \nabla F_n(\theta_{B,t})$ 
4:    $\tilde{B} \leftarrow \tilde{B} - nC_{\text{grad}}$ 
5: end if
6: if  $\tilde{B} \geq nC_{\text{eval}}$  then
7:   Let  $g_t = F_n(\theta_{B,t})$ 
8:    $\tilde{B} \leftarrow \tilde{B} - nC_{\text{eval}}$ 
9: end if
10: while  $\|G_t\| > \tau$  and  $\tilde{B} \geq nC_{\text{eval}}$  do
11:   Let  $g'_t = F_n(\theta_{B,t} - G_t)$  and  $v_{t+1} = 1$ 
12:    $\tilde{B} \leftarrow \tilde{B} - nC_{\text{eval}}$ 
13:   while  $g'_t > g_t - \frac{v_{t+1}}{2}\|G_t\|^2$  and  $\tilde{B} \geq nC_{\text{eval}}$  do
14:      $v_{t+1} \leftarrow \beta v_{t+1}$ 
15:      $g'_t \leftarrow F_n(\theta_{B,t} - v_{t+1}G_t)$ 
16:      $\tilde{B} \leftarrow \tilde{B} - nC_{\text{eval}}$ 
17:   end while
18:   if  $g'_t \leq g_t - \frac{v_{t+1}}{2}\|G_t\|^2$  then
19:     Let  $\theta_{B,t+1} = \theta_{B,t} - v_{t+1}G_t$  and  $g_{t+1} = g'_t$ 
20:      $t \leftarrow t + 1$ 
21:     if  $\tilde{B} \geq nC_{\text{grad}}$  then
22:       Let  $G_t \leftarrow \nabla F_n(\theta_{B,t})$ 
23:        $\tilde{B} \leftarrow \tilde{B} - nC_{\text{grad}}$ 
24:     else
25:       break
26:     end if
27:   end if
28: end while
29: return:  $(\theta_{B,t}, \tilde{B} \vee 0)$ 

```

Theorem 1. Assume that A1–A2 hold. Let $\theta_0 \in \mathbb{R}^d$, $\beta \in (0, 1)$ and $n : \mathbb{N} \rightarrow \mathbb{N}$ be such that $\lim_{B \rightarrow \infty} n(B)/B = 0$ and such that $\lim_{B \rightarrow \infty} n(B) = \infty$. In addition, for all $B \in \mathbb{N}$ and with $\alpha \in (0, 1]$ as in A2 let

$$(\tilde{\theta}_B, \tilde{B}) = \text{Algorithm 3}(\theta_0, n(B), B, 0, \beta), \quad r(B) = \max(\sqrt{n(B)/B}, n(B)^{-\frac{\alpha}{1+\alpha}}).$$

Then, $\|\tilde{\theta}_B - \theta_\star\| = \mathcal{O}_{\mathbb{P}}(r(B))$ and $\liminf_{B \rightarrow \infty} B^{\frac{\alpha}{1+3\alpha}} r(B) > 0$.

The expression for the rate $r(B)$ has a very intuitive and simple explanation. Letting Θ be a compact set containing $\theta_\star$, we have

$$(3) \quad \|\tilde{\theta}_B - \theta_\star\| \leq \|\tilde{\theta}_B - \hat{\theta}_{\Theta, n(B)}\| + \|\hat{\theta}_{\Theta, n(B)} - \theta_\star\|,$$

where the first term on the right-hand side is the numerical error resulting from using GD-BLS to minimize $F_{n(B)}$. Under the assumptions of Theorem 1, if T iterations of GD-BLS are performed then the numerical error is of size $T^{-1/2}$.

In the proof of the theorem we show that, under A1–A2, the cost per iteration of Algorithm 3 is approximatively constant, implying that $\tilde{\theta}_B$ is computed by running $T \approx B/n(B)$ iterations of GD-BLS. Consequently, the numerical error $\|\tilde{\theta}_B - \hat{\theta}_{\Theta, n(B)}\|$ is of size $\sqrt{B/n(B)}$, which is the first term appearing in the definition of $r(B)$. The second term in (3) is the statistical error which, by Lemma 5, is of size $\mathcal{O}_{\mathbb{P}}(n(B)^{-\frac{\alpha}{1+\alpha}})$. Therefore, the rate $r(B)$ given in Theorem 1 is simply the maximum between the numerical error and the statistical error.

The last part of Theorem 1 follows from the fact that the fastest rate at which $r(B)$ can decrease to zero is obtained when the numerical error and the statistical error vanish at the same speed. It is direct to see that this happens when $n(B) \approx B^{\frac{1+\alpha}{1+3\alpha}}$, in which case $r(B) = \mathcal{O}(B^{-\frac{\alpha}{1+3\alpha}})$. This result is formalized in Corollary 2.

Corollary 2. *Consider the set-up of Theorem 1 with $n : \mathbb{N} \rightarrow \mathbb{N}$ such that*

$$0 < \liminf_{B \rightarrow \infty} \frac{n(B)}{B^{\frac{1+\alpha}{1+3\alpha}}} \leq \limsup_{B \rightarrow \infty} \frac{n(B)}{B^{\frac{1+\alpha}{1+3\alpha}}} < \infty.$$

Then, $\|\tilde{\theta}_B - \theta_\star\| = \mathcal{O}_{\mathbb{P}}(n(B)^{-\frac{\alpha}{1+\alpha}}) = \mathcal{O}_{\mathbb{P}}(B^{-\frac{\alpha}{1+3\alpha}})$.

Algorithm 3 belongs to the well-known class of sample average approximation algorithms (see [10] for a survey of sample average approximation methods), which aim at computing an estimate of $\theta_\star$ by minimizing the function $F_{n(B)}$ with a deterministic optimization routine $\mathcal{R}$. The optimal allocation of the available computational resources (i.e. of the optimal choice of $n(B)$) is studied in [17] under general conditions on the optimizer $\mathcal{R}$ but, unlike Theorem 1 and Corollary 2, the results presented in this reference assume that the search space is compact and they are limited to the case where $\alpha = 1$.

3.2. An improved application of gradient descent to noisy optimization.

3.2.1. The algorithm. The procedure sketched in the introductory section to improve upon the convergence rate of $\tilde{\theta}_B$ obtained in Corollary 2 is presented in Algorithm 4. Before providing, in Theorem 2, a convergence result for this algorithm some comments are in order.

Firstly, in Algorithm 4, for all j a specific form is imposed on the number $n_j(B)$ of elements of $(Z_i)_{i \geq 1}$ and on the tolerance parameter $\tau_j(B)$ that are used in the j -th call of Algorithm 3. This is only to simplify the presentation of the algorithm and the conclusions of Theorem 2 hold for any alternative functions $\tilde{n}_j$ and $\tilde{\tau}_j$ satisfying

$$0 < \liminf_{B \rightarrow \infty} \frac{n_j(B)}{\tilde{n}_j(B)} \leq \limsup_{B \rightarrow \infty} \frac{n_j(B)}{\tilde{n}_j(B)} < \infty, \quad 0 < \liminf_{B \rightarrow \infty} \frac{\tau_j(B)}{\tilde{\tau}_j(B)} \leq \limsup_{B \rightarrow \infty} \frac{\tau_j(B)}{\tilde{\tau}_j(B)} < \infty.$$

Secondly, in Algorithm 4 we have $\tau_j(B) > 0$ even when $j = J$, that is even in the last call of Algorithm 3 the optimization of the function $F_{n_j(B)}$ is stopped earlier when its gradient is small. However, there is no reason for stopping earlier the minimization of $F_{n_J(B)}$ and our convergence result (Theorem 2) remains valid when, for $j = J$, line 5 of Algorithm 4 is replaced by

$$5 : \quad (\hat{\theta}_{j,B}, B_j) = \text{Algorithm 3}(\hat{\theta}_{j-1,B}, \lceil \kappa B^{\gamma_j} \rceil, B_{j-1}, 0, \beta).$$

Nevertheless, in its current form, Algorithm 4 has the following advantage: if we let $J = J_1$ for some $J_1 > 1$ but, for some $J_2 \in \{1, \dots, J_1 - 1\}$, the computational

Algorithm 4

Input: Starting value $\theta_0 \in \mathbb{R}^d$, computational budget $B \in \mathbb{N}$, rate parameters $\alpha' \in (0, 1]$ and $\delta \in [0, 1)$, maximum number of GD-BLS calls $J \in \mathbb{N}$, sample size parameters $\kappa \in (0, \infty)$, tolerance parameter $\tau \in (0, \infty)$ and backtracking parameter $\beta \in (0, 1)$.

- 1: Let $j = 0$, $B_0 = B$ and $\hat{\theta}_{0,B} = \theta_0$
- 2: **while** $B_j > 0$ and $j < J$ **do**
- 3: Let $j = j + 1$ and $\gamma_j = 1 - \delta^j$
- 4: Let $(\hat{\theta}_{j,B}, B_j) = \text{Algorithm 3}(\hat{\theta}_{j-1,B}, \lceil \kappa B^{\gamma_j} \rceil, B_{j-1}, \tau B^{-\frac{\alpha'}{1+\alpha'}\gamma_j}, \beta)$
- 5: **end while**
- 6: **return:** $\hat{\theta}_{j,B}$

budget is exhausted after the J_2 -th call of Algorithm 3 then the estimate of θ_* returned by Algorithm 4 is the one we would have obtained if we had chosen $J = J_2$. From a practical point of view, this means that we can choose a large value for J and let the algorithm determines how many calls to Algorithm 3 can be made given the available computational resources.

3.2.2. Convergence result. In fact, expanding on the latter point, the only reason why Algorithm 4 imposes an upper bound J on the number of calls of Algorithm 3 is to prove the following result:

Theorem 2. Assume that A1–A2 hold. Let $\alpha' \in (0, 1]$, $\theta_0 \in \mathbb{R}^d$, $J \in \mathbb{N}$, $(\kappa, \tau) \in (0, \infty)^2$, $\beta \in (0, 1)$, $\delta \in (2\alpha'/(1+3\alpha'), 1)$ and

$$(\hat{\theta}_{J,B}, \tilde{B}) = \text{Algorithm 4}(\theta_0, B, \alpha', \delta, J, \kappa, \tau, \beta), \quad \forall B \in \mathbb{N}.$$

Then, $\|\hat{\theta}_{J,B} - \theta_*\| = o_{\mathbb{P}}(1)$ and if A2 holds for $\alpha = \alpha'$ then

$$\|\hat{\theta}_{J,B} - \theta_*\| = \mathcal{O}_{\mathbb{P}}\left(B^{-\frac{\alpha'}{1+\alpha'}(1-\delta^J)}\right).$$

Assuming that A2 holds for $\alpha = \alpha'$, it follows from Theorem 2 that taking a small value for J can only make slower the speed at which $\hat{\theta}_{J,B}$ converges to θ_* . However, as mentioned in the introductory section, to observe in practice the $\mathcal{O}(B^{-\frac{\alpha'}{1+\alpha'}(1-\delta^J)})$ convergence rate it is needed to minimize the function $F_{n_J(B)}$, which may not be the case if J is too large given the available computational budget. For this reason, we expect the observed convergence rate to be $B^{-\frac{\alpha'}{1+\alpha'}(1-\delta^{J_B})}$, with J_B the number of calls of Algorithm 3 needed for computing $\hat{\theta}_{J,B}$, that is with J_B the smallest value of $j \in \{1, \dots, J\}$ such that $\hat{\theta}_{j,B} = \hat{\theta}_{J,B}$. A sensible approach is therefore to let the algorithm decide how many calls to Algorithm 3 are needed by choosing J large enough so that $J_B < J$ with probability close to one. The numerical experiments presented in the next section suggest that J_B increases with B at speed $\log(B)$, and verifying theoretically this observation is left for future research.

Noting that, for any $j \geq 1$, increasing δ makes the j -th call of Algorithm 3 computationally less expensive, it follows that J_B increases with δ . Hence, the larger δ the slower the convergence rate of $\hat{\theta}_{j,B}$ for any given $j \geq 1$, but the larger

the number of calls to Algorithm 3 can be made. The numerical experiments of Section 4 suggest that the choice of δ has little impact on the convergence behaviour of $\hat{\theta}_{J,B}$.

Finally, if A2 does not hold for $\alpha = \alpha'$ then the proof of Theorem 2 shows that $\|\hat{\theta}_{J,B} - \theta_\star\|$ converges to zero at least at speed $B^{-\frac{\alpha}{1+\alpha}(1-\delta)}$, for any α such that A2 holds. Studying more precisely the convergence rate of $\hat{\theta}_{J,B}$ in this context is left for future research, but we expect this rate to be slower than the one obtained when A2 holds for the chosen value of α' . To understand why, remark that in Algorithm 4 the parameter α' only impacts the value of the tolerance parameter used in each call of Algorithm 3, and that the larger α' the smaller the tolerance parameter. Then, if α' is too large (i.e. if A2 does not hold for $\alpha = \alpha'$), for any $j \in \{1, \dots, J\}$ too much computational budget is allocated to the j -th call of Algorithm 3, resulting in a waste of computational resources and thus in a slower convergence rate as $B \rightarrow \infty$. In particular, it may be the case that even for large values of B we have $J_B < J$ with high probability.

3.2.3. Understanding where the fast convergence rate comes from. To understand why Algorithm 4 allows to significantly improve upon the convergence rate obtained in Section 3.1, consider the set-up of Theorem 1 with the fixed starting value θ_0 replaced by a sequence $(\theta_{0,B})_{B \geq 1}$ of starting values such that $\|\theta_{0,B} - \theta_\star\| = \mathcal{O}(B^{-c})$ for some constant $c \geq 0$, that is let

$$(\tilde{\theta}_{c,B}, \tilde{B}) = \text{Algorithm 3}(\theta_{0,B}, n_c(B), B, 0, \beta), \quad \forall B \in \mathbb{N}$$

for some function n_c verifying the assumptions of Theorem 1. In this case, our computations show that

$$(4) \quad \|\tilde{\theta}_{c,B} - \theta_\star\| = \mathcal{O}_{\mathbb{P}}(r_c(B)), \quad r_c(B) = \max \left(B^{-c} \sqrt{n_c(B)/B}, n_c(B)^{-\frac{\alpha}{1+\alpha}} \right),$$

where, as in Section 3.1 and for some $\Theta \subset \mathbb{R}^d$, the first term in the definition of $r_c(B)$ is the numerical error $\|\tilde{\theta}_{c,B} - \hat{\theta}_{\Theta, n_c(B)}\|$ and the second one is the statistical error $\|\hat{\theta}_{\Theta, n_c(B)} - \theta_\star\|$. From (4) we observe that the value of c modifies the trade-off computational error versus statistical error we are facing when choosing $n_c(B)$, and these two sources of errors converge to zero at the same speed when

$$(5) \quad 0 < \liminf_{B \rightarrow \infty} \frac{n_c(B)}{B^{\frac{(1+\alpha)(1+2c)}{1+3\alpha}}} \leq \limsup_{B \rightarrow \infty} \frac{n_c(B)}{B^{\frac{(1+\alpha)(1+2c)}{1+3\alpha}}} < \infty$$

in which case

$$\|\tilde{\theta}_{c,B} - \theta_\star\| = \mathcal{O}_{\mathbb{P}} \left(B^{-\frac{\alpha(1+2c)}{1+3\alpha}} \right).$$

Therefore, initializing Algorithm 3 with a starting value that converges to $\theta_\star$ (i.e. having $c > 0$) allows to improve the rate at which we learn $\theta_\star$. In particular, for $c = 0$ we recover the result of Corollary 2, and the larger c the faster the convergence rate of $\tilde{\theta}_{c,B}$.

3.2.4. Understanding the convergence rate given in Theorem 2. The convergence rate given in Theorem 2 is more easily understood by studying an alternative (but less practical, see below) estimator of $\theta_\star$.

To define this estimator let $\delta_{\alpha'} = 2\alpha'/(1+3\alpha')$, $c_j = (1 - \delta_{\alpha'}^j)\alpha'/(1+\alpha')$ for all $j \geq 0$ and let $\theta_0 \in \mathbb{R}^d$. In addition, let $J \in \mathbb{N}$ and $\{\kappa_{J,j}\}_{j=1}^J$ be such that $\sum_{j=1}^J \kappa_{J,j} = 1$ and such that $\kappa_{J,j} > 0$ for all $j \in \{1, \dots, J\}$. Finally, for all

$j \in \{1, \dots, J\}$ and using the convention that $\tilde{\theta}_{j-1,B,J} = \theta_0$ for all $B \in \mathbb{N}$ when $j = 1$, let

$$(\tilde{\theta}_{j,B,J}, \tilde{B}_j) = \text{Algorithm 3} \left(\tilde{\theta}_{j-1,B,J}, n_{c_{j-1}}(B), \lfloor \kappa_{J,j} B \rfloor, 0, \beta \right), \quad \forall B \in \mathbb{N}$$

with the function $n_{c_{j-1}}$ such that (5) holds (with $c = c_{j-1}$). Then, assuming that A2 holds for $\alpha = \alpha'$ and using the computations of Section 3.2.3, it is easily checked that for all $j \in \{1, \dots, J\}$ we have $\|\tilde{\theta}_{j,B,J} - \theta_\star\| = \mathcal{O}_{\mathbb{P}}(B^{-c_j})$, where

$$B^{-c_j} = \max \left(B^{-c_{j-1}} \sqrt{n_{c_{j-1}}(B)/B}, n_{c_{j-1}}(B)^{-\frac{\alpha'}{1+\alpha'}} \right), \quad \forall B \in \mathbb{N}.$$

Informally speaking, and following the reasoning of Section 3.2.3, this implies that $\tilde{\theta}_{j,B,J}$ is computed from $\tilde{\theta}_{j-1,B,J}$ using the number $n_{c_{j-1}}(B)$ of elements of $(Z_i)_{i \geq 1}$ that maximizes the speed at which $\|\tilde{\theta}_{j,B,J} - \theta_\star\|$ converges to zero. Using the definition of $(c_j)_{j \geq 1}$, it is readily checked that, for all $j \in \{1, \dots, J\}$,

$$\|\tilde{\theta}_{j,B,J} - \theta_\star\| = \mathcal{O}_{\mathbb{P}} \left(B^{-\frac{\alpha'}{1+\alpha'}(1-\delta_j^j)} \right)$$

while Theorem 2 establishes that $\|\hat{\theta}_{j,B} - \theta_\star\| = \mathcal{O}_{\mathbb{P}}(B^{-\frac{\alpha'}{1+\alpha'}(1-\delta^i)})$ for $\delta \in (\delta_{\alpha'}, 1)$.

In addition to giving some insight on this latter result, the above calculations show that $\tilde{\theta}_{j,B,J}$ converges to $\theta_\star$ (slightly) faster than $\hat{\theta}_{j,B}$ for any $j \in \{1, \dots, J\}$. However, these alternative estimators have the important drawback to require that the computational budget allocated to each call of Algorithm 3 is chosen beforehand, and in a way that depends on the chosen value of J . By contrast, in Algorithm 4 this allocation is done automatically and does not depend on J , and if only $j < J$ calls to Algorithm 3 are made then the algorithm returns the estimate of $\theta_\star$ we would have obtained if we had chosen $J = j$. In our calculations, these desirable properties of Algorithm 4 are obtained at the price of imposing that δ is strictly bigger than $\delta_{\alpha'}$.

3.3. Validity of GD-BLS for solving noisy optimization problems: Sketch of proof. We end this section with a sketch of proof of the validity of GD-BLS for solving noisy optimization problems, the goal being to provide the reader with some insights about how Assumptions A1–A2 and the properties of the GD-BLS algorithm are used to prove our main results. Specifically, in this subsection we let $\theta_0 \in \mathbb{R}^d$ and consider the problem of solving the optimization problem (1) by applying the GD-BLS algorithm (i.e. Algorithms 1–2) on the random function F_n , using θ_0 as starting value. We denote by $(\theta_t^{(n)})_{t \geq 1}$ the resulting sequence of iterates and below we show, in an informal way, that the results of Sections 2.3–2.4 imply that $\|\lim_{t \rightarrow \infty} \theta_t^{(n)} - \theta_\star\| = o_{\mathbb{P}}(1)$.

To this aim remark first that, by using the descent property of GD-BLS, all the elements of $(\theta_t^{(n)})_{t \geq 1}$ belong to the (random) set $\tilde{\Theta}_{0,n} := \{\theta \in \mathbb{R}^d : F_n(\theta) \leq F_n(\theta_0)\}$. Next, by using Lemma 3 and under A1, it can be proved that there exists a compact and convex set $\tilde{\Theta}_0$ such that $\tilde{\Theta}_{0,n} \subset \tilde{\Theta}_0$ with $\mathbb{P}$ -probability tending to one. To proceed further we let

$$\Theta_{0,n} = \left\{ \theta \in \mathbb{R}^d : \exists \theta' \in \tilde{\Theta}_0 \text{ such that } \|\theta - \theta'\| \leq \sup_{\tilde{\theta} \in \tilde{\Theta}_0} \|\nabla F_n(\tilde{\theta})\| \right\},$$

$$\Theta_0 = \left\{ \theta \in \mathbb{R}^d : \exists \theta' \in \tilde{\Theta}_0 \text{ such that } \|\theta - \theta'\| \leq \sup_{\tilde{\theta} \in \tilde{\Theta}_0} \|\nabla F(\tilde{\theta})\| + 1 \right\}$$

and remark that, by Lemma 3, with $\mathbb{P}$ -probability tending to one we have $\Theta_{0,n} \subset \Theta_0$. Under A1 the set Θ_0 is compact and convex and thus, by Lemma 1 and Corollary 1, it follows that with $\mathbb{P}$ -probability tending to one the function F_n is strictly convex and L -smooth on Θ_0 (with $L \in (0, \infty)$ a finite constant). Without loss of generality we can assume that Θ_0 is sufficiently large so that θ_* belongs to the interior of this set, in which case $\|\hat{\theta}_{\Theta_{0,n}} - \theta_*\| = o_{\mathbb{P}}(1)$ by Lemma 5. To sum-up, with $\mathbb{P}$ -probability tending to one (i) the function F_n is strictly convex and L -smooth on the compact and convex set Θ_0 , (ii) $F_n(\hat{\theta}_{\Theta_{0,n}}) = \inf_{\theta \in \Theta_0} F_n(\theta)$ and (iii) the sequence of iterates $(\theta_t^{(n)})_{t \geq 1}$ generated by the GD-BLS algorithm applied on F_n is a sequence in $\hat{\Theta}_0$. Then, by using Propositions 5–6, it readily follows that $\|\lim_{t \rightarrow \infty} \theta_t^{(n)} - \hat{\theta}_{\Theta_{0,n}}\| = o_{\mathbb{P}}(1)$, and since $\|\hat{\theta}_{\Theta_{0,n}} - \theta_*\| = o_{\mathbb{P}}(1)$ we can conclude that $\|\lim_{t \rightarrow \infty} \theta_t^{(n)} - \theta_*\| = o_{\mathbb{P}}(1)$.

4. NUMERICAL EXPERIMENTS

In this section we study the behaviour of $\mathbb{E}[\|\hat{\theta}_{J,B} - \theta_*\|]$ and of $\mathbb{E}[J_B]$ as B increases, as estimated from 100 independent realizations of $(Z_i)_{i \geq 1}$. In all the experiments presented below, Algorithm 4 is implemented with $\beta = 0.5$ and with $C_{\text{grad}} = C_{\text{eval}} = \kappa = \tau = 1$, and with $\lceil \kappa B^{\gamma_j} \rceil$ replaced by $\max(100, \lceil \kappa B^{\gamma_j} \rceil)$ for all $j \geq 1$. It is trivial that Theorem 2 remains valid with this modification of the algorithm, which is introduced to prevent to minimize $F_{n_j(B)}$ for a value of $n_j(B)$ that is too small. In addition, following the discussion of Section 3.2, we let $J = 10^4$ so that $J_B < J$ with probability close to one. Finally, in what follows we denote by ρ_{J_B} the coefficient of correlation between the variables $(\mathbb{E}[J_B])_{B \geq 1}$ and $(\log(B))_{B \geq 1}$, as estimated from 100 independent realizations of $(Z_i)_{i \geq 1}$. Remark that a value of ρ_{J_B} close to one implies that $\mathbb{E}[J_B] \approx a + b \log(B)$ for some constants $a \in \mathbb{R}$ and $b \in (0, \infty)$.

4.1. Example of Proposition 1. We first consider the univariate optimization problem defined in Proposition 1, which we recall amounts to solving the $d = 1$ dimensional noisy optimization problem (1) with $Z = (X, Y)$, where X and Y are two independent Poisson random variables such that $\mathbb{E}[X] = \mathbb{E}[Y] = 1$, and with the function $f : \mathbb{R}^d \times \mathbb{Z} \rightarrow \mathbb{R}$ defined by

$$f(\theta, z) = -z_2 z_1 \theta + \exp(\theta z_1), \quad \theta \in \mathbb{R}, \quad z \in \mathbb{Z} := \mathbb{R}^2.$$

As shown in Proposition 3, for this optimization problem Assumptions A1–A2 hold with $\alpha = 1$ and, to the best of our knowledge, SG is not guaranteed to converge when used to solve it (see Propositions 1–2). Simple computations show that the global minimizer of the resulting objective function F is $\theta_* = 0$. For this example we let $\theta_0 = 1$ and $\alpha' = 1$, in which case Theorem 2 imposes to have $\delta \in (0.5, 1)$.

Figure 1 shows, for $\delta = 0.51$ and for $\delta = 0.95$, the evolution of $\mathbb{E}[\|\hat{\theta}_{J,B} - \theta_*\|]$ and of $\mathbb{E}[J_B]$ as B increases. For the two considered values of δ the observed rate at which $\mathbb{E}[\|\hat{\theta}_{J,B} - \theta_*\|]$ decreases to zero is indistinguishable from the optimal $B^{-1/2}$ convergence rate. In addition, in this experiment the value of δ appears to have little impact on $\mathbb{E}[\|\hat{\theta}_{J,B} - \theta_*\|]$. As expected, we see in Figure 1 that the number of calls to Algorithm 3 is much larger for $\delta = 0.95$ than for $\delta = 0.51$, with a value of $\mathbb{E}[J_B]$ ranging from 20.48 to 34.04 for the former choice of δ and ranging from 2.37 to 3.28 for the latter. Visually, we observe (at least for $\delta = 0.95$) that $\mathbb{E}[J_B]$ grows with B as $\log(B)$. The coefficient of correlation between the variables

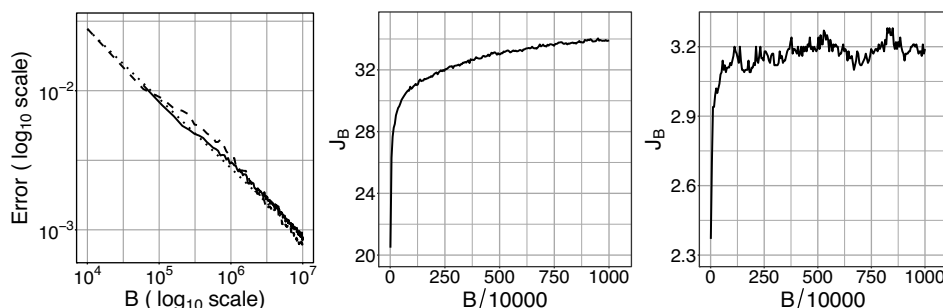


FIGURE 1. Results for the example of Section 4.1. The left plot shows $\mathbb{E}[\|\hat{\theta}_{J,B} - \theta_\star\|]$ as a function of B , where the solid line is for $\delta = 0.95$ and the dashed line for $\delta = 0.51$, and with the two dotted lines representing the $B^{-1/2}$ convergence rate. The middle and right plots show the evolution of $\mathbb{E}[J_B]$ as a function of B for $\delta = 0.95$ (middle plot) and for $\delta = 0.51$ (right plot). All the results are obtained from 100 independent realizations of $(Z_i)_{i \geq 1}$.

$(\mathbb{E}[J_B])_{B \geq 1}$ and $(\log(B))_{B \geq 1}$ confirms this observation, since for this experiment we have $\rho_{J_B} = 0.99$ when $\delta = 0.95$ and $\rho_{J_B} = 0.78$ when $\delta = 0.51$.

4.2. A multivariate extension of the example of Proposition 1. We now consider the $d = 20$ dimensional extension of the example of Proposition 1 obtained by letting, in (1), $Z = (W, X, Y)$ be such that

$$W \sim \text{Poisson}(1), \quad X|W \sim \mathcal{U}([-1, 1]^{d-1}), \quad Y|(W, X) \sim \text{Poisson}(a_\star^\top X)$$

for some $a_\star \in \mathbb{R}^{d-1}$, and the function $f : \mathbb{R}^d \times Z \rightarrow \mathbb{R}$ be defined by

$$f(\theta, z) = -z_2 \theta^\top z_1 + \exp(\theta^\top z_1), \quad z = (z_1, z_2), \quad \theta \in \mathbb{R}^d, \quad z_1 \in \mathbb{R}^d, \quad z_2 \in \mathbb{R}.$$

It is easily checked that the resulting objective function F is such that A1–A2 hold with $\alpha = 1$ and that $\theta_\star = (0, a_\star)$. Compared to the example of Proposition 1, we expect that by increasing the dimension of the search space we make the optimization problem more challenging and thus that the conclusions of Propositions 1–2 hold for this example. For this experiment the value $a_\star$ is a random draw from the $\mathcal{N}_{d-1}(0, \mathbb{1}_{d-1})$ distribution and, as in the previous subsection, the results are presented for $(\alpha', \delta) = (1, 0.51)$ and for $(\alpha', \delta) = (1, 0.95)$. Finally, Algorithm 4 is implemented using $\theta_0 = (1, \dots, 1)$ as starting value.

From Figure 2, which shows the evolution of $\mathbb{E}[\|\hat{\theta}_{J,B} - \theta_\star\|]$ and of $\mathbb{E}[J_B]$ as B increases, we see that the behaviour of these two quantities is similar to what we observed in the previous subsection. The main difference is however that, because of the larger dimension of θ , the $B^{-1/2}$ convergence rate for $\hat{\theta}_{J,B}$ is observed only for B large enough (roughly for $B > 10^{6.5}$). The study of the dependence of the performance of Algorithm 4 on the dimension d of θ is left for future research. Finally, in Figure 2 we also observe that $\mathbb{E}[J_B]$ appears to increase with B as $\log(B)$, and indeed, for this example, the coefficient of correlation between $(\mathbb{E}[J_B])_{B \geq 1}$ and $(\log(B))_{B \geq 1}$ is $\rho_{J_B} = 0.98$ for $\delta = 0.95$ and $\rho_{J_B} = 0.70$ for $\delta = 0.51$.

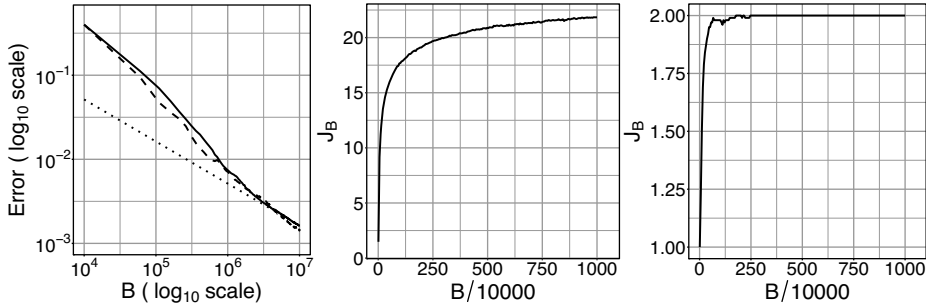


FIGURE 2. Results for the example of Section 4.2. The left plot shows $\mathbb{E}[\|\hat{\theta}_{J,B} - \theta_\star\|]$ as a function of B , where the solid line is for $\delta = 0.95$ and the dashed line for $\delta = 0.51$, and with the dotted line representing the $B^{-1/2}$ convergence rate. The middle and right plots show the evolution of $\mathbb{E}[J_B]$ as a function of B for $\delta = 0.95$ (middle plot) and for $\delta = 0.51$ (right plot). All the results are obtained from 100 independent realizations of $(Z_i)_{i \geq 1}$

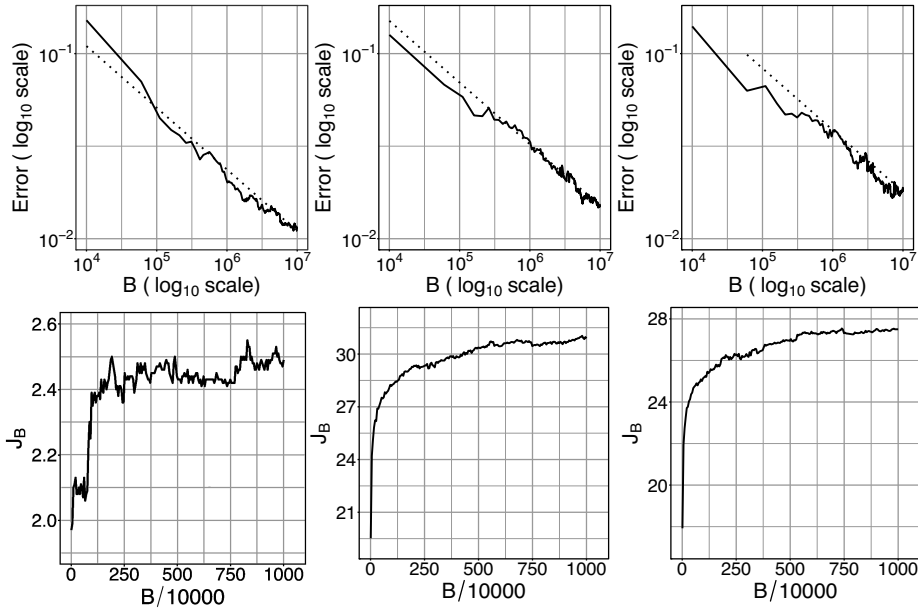


FIGURE 3. Results for the example of Section 4.3. The top plots show $\mathbb{E}[\|\hat{\theta}_{J,B} - \theta_\star\|_{0.1}]$ as a function of B and the bottom plots show $\mathbb{E}[J_B]$ as a function of B . The left plots are for $(\alpha', \delta) = (0, 5, 0.41)$, the middle plots for $(\alpha', \delta) = (0.5, 0.95)$ and the right plots for $(\alpha', \delta) = (1, 0.95)$. In the top plots the dotted lines represent the $B^{-1/3}$ convergence rate and all the results are obtained from 100 independent realizations of $(Z_i)_{i \geq 1}$

4.3. Example of Proposition 1 with an extra heavy tailed noise. We finally consider the example of Proposition 1 with an extra heavy tailed noise. More

precisely, we aim at solving the $d = 1$ noisy optimization problem (1) with $Z = (W, X, Y)$, where $W \sim t_\nu(0, 1)$ and where X and Y are two independent Poisson random variables which are independent of W and such that $\mathbb{E}[X] = \mathbb{E}[Y] = 1$, and with the function $f : \mathbb{R}^d \times \mathbf{Z} \rightarrow \mathbb{R}$ defined by

$$f(\theta, z) = -z_3 z_2 \theta + \exp(\theta z_2) + z_1 \theta, \quad \theta \in \mathbb{R}, \quad z \in \mathbf{Z} := \mathbb{R}^3.$$

We let $\nu = 1.501$ so that A2 holds for any $\alpha < \nu - 1 = 0.501$. Remark that since $\mathbb{E}[W] = 0$ it follows that the resulting objective function F is the same as for the example of Proposition 1, and thus $\theta_\star = 0$. It is also immediate to see that the conclusions of Propositions 1–2 remain valid for this example, and therefore to be best of our knowledge SG is not guaranteed to converge when applied to solve it. In addition, the results of [14] for clipped SG do not apply to this optimization problem, notably because the function F is not L -smooth (see Proposition 1). For this example we implement Algorithm 4 with $\theta_0 = 1$ and the results are presented for $\alpha' = 0.5$ and for $\alpha' = 1$. Remark that A2 holds with $\alpha = \alpha'$ for the former value of α' but not for the latter. Remark also that, by Theorem 2, we must have $\delta \in (0.4, 1)$ when $\alpha' = 0.5$ and $\delta \in (0.5, 1)$ when $\alpha' = 1$.

Due to the heavy tailed noise, for this example it is not guaranteed that we have $\mathbb{E}[\|\hat{\theta}_{J,B}\|] < \infty$ and the numerical experiments suggest that this may indeed not be the case. For this reason, we study, in Figure 3, the behaviour as B increases of $\mathbb{E}[\|\hat{\theta}_{J,B} - \theta_\star\|_{0.1}]$, the expected 10% trimmed mean of $\|\hat{\theta}_{J,B} - \theta_\star\|$.

The plots in the first two columns of Figure 3 are obtained for $(\alpha', \delta) = (0.5, 0.41)$ and for $(\alpha', \delta) = (0.5, 0.95)$. From these plots we see that the convergence rate of $\hat{\theta}_{J,B}$ is undistinguishable from the $B^{-\frac{\alpha'}{1+\alpha'}} = B^{-1/3}$ convergence rate, confirming the result of Theorem 2. As for the examples of Sections 4.1–4.2, the choice of δ appears to have only a limited impact on the estimation error. We also observe that for the two considered values of δ the quantity $\mathbb{E}[J_B]$ seems to grow with B as $\log(B)$, an observation which is confirmed by the fact that $\rho_{J_B} = 0.85$ for $\delta = 0.41$ and $\rho_{J_B} = 0.98$ for $\delta = 0.95$. The plots in last column of Figure 3 study the behaviour of $\hat{\theta}_{J,B}$ when $(\alpha', \delta) = (1, 0.95)$, in which case A2 does not hold for $\alpha = \alpha'$. If $\hat{\theta}_{J,B}$ appears to roughly converge to $\theta_\star$ at speed $B^{-1/3}$, we observe that the convergence becomes slower for values of B greater than $10^{6.7}$. Remark that, from the discussion of Section 3.2.2, it was expected that the convergence rate of $\hat{\theta}_{J,B}$ is slower when A2 does not hold for the chosen value of α' than when it does. Finally, for this value of (α', δ) the quantity $\mathbb{E}[J_B]$ also seems to grow as $\log(B)$, an observation which is confirmed by the large value $\rho_{J_B} = 0.98$ obtained for the coefficient of correlation between $(\mathbb{E}[J_B])_{B \geq 1}$ and $(\log(B))_{B \geq 1}$.

5. CONCLUDING REMARKS

In this paper we show, theoretically and empirically, that gradient descent with backtracking line search can be used to solve noisy optimization problems with a nearly optimal convergence rate, and without assuming that the objective function is globally L -smooth.

The proposed algorithm (Algorithm 4) requires that GD-BLS can be called at most J times, with J a user-specified parameter. We conjecture that this condition can be removed and, assuming that A2 holds for $\alpha = \alpha'$, that the resulting estimator $\hat{\theta}_B$ of the global minimizer $\theta_\star$ converges at speed $B^{-\frac{\alpha'}{1+\alpha'}(1-j_B)}$ for some sequence

$(j_B)_{B \geq 1}$ such that $\lim_{B \rightarrow \infty} j_B = \infty$. Based on our numerical experiments, we predict that $j_B = a + b \log(B)$ for some constants $a \in [0, \infty)$ and $b \in (0, \infty)$ depending on the choice of δ . Confirming these results theoretically is left for future research.

Another interesting question that deserves further work is the possibility to replace, in Algorithm 4, GD-BLS by GD-BLS with Nesterov momentum (GD-BLS-NM). We recall the reader that, while the convergence of the former algorithm for computing $\min_{x \in \mathbb{R}^d} g(x)$ for a convex and L -smooth function g is $1/T$, with T the number of iterations, GD-BLS-NM converges to the global minimum of g at the faster (and optimal) $1/T^2$ rate (see Section 2.2 in [12]). Based on this result, it is reasonable to expect that replacing GD-BLS by GD-BLS-NM will improve the non-asymptotic behaviour of the proposed procedure for computing θ_* . The extension of our convergence results to the case where GD-BLS-NM is used is however non-trivial, notably because it is unclear if the aforementioned fast convergence rate of this algorithm remains valid when the function g is only locally L -smooth. The $1/T$ convergence rate of GD-BLS holds for such functions because the backtracking procedure ensures that the estimate of $\min_{x \in \mathbb{R}^d} g(x)$ improves from one iteration to the next, and thus that the sequence in $\mathbb{R}^d$ generated by the algorithm stays in a compact set. By contrast, GD-BLS-NM is not a descent algorithm, and proving that the sequence in $\mathbb{R}^d$ it generates remains in a compact set is non-trivial.

Finally, it is worth mentioning that, instead of backtracking line search, the procedure introduced by [11] can be used for choosing step-sizes $(v_t)_{t \geq 1}$ which guarantee that the resulting gradient descent algorithm generates a sequence in $\mathbb{R}^d$ that stays in a compact set, when the objective function is convex. Under the additional assumption that the function $f(\cdot, Z)$ is $\mathbb{P}$ -a.s. convex, it is easily checked that all the theoretical results presented in this work are valid if, instead of performing backtracking line search, in Algorithm 3 the step-sizes $(v_t)_{t \geq 1}$ are chosen as proposed by [11]. The key advantage of this approach is that, in the context of this work, it does not consume any computational budget and therefore allows to reduce the cost per iteration of Algorithm 3. However, this procedure has the drawback of requiring to choose the initial element v_1 of the sequence of $(v_t)_{t \geq 1}$ which usually (and as confirmed by some unreported experiments) has a large impact on the behaviour of the algorithm.

APPENDIX A. PROOFS

A.1. Proof of Proposition 2.

Proof of Proposition 2. For all $\theta \in \mathbb{R}$ we have

$$\begin{aligned} F(\theta) &= -\theta + \exp(\exp(\theta) - 1), \\ (6) \quad \nabla F(\theta) &= -1 + \exp(\exp(\theta) + \theta - 1), \\ \nabla^2 F(\theta) &= (\exp(\theta) + 1) \exp(\exp(\theta) + \theta - 1). \end{aligned}$$

We now show that there exists no $\alpha \in (0, 1]$ such that Assumption 5 of [16] holds. To this aim, for all $\alpha \in (0, 1]$ we let

$$(7) \quad G_\alpha(\theta) = \mathbb{E}[|\nabla f(\theta, Z)|^{1+\alpha}], \quad \forall \theta \in \mathbb{R}$$

and, for all $\epsilon > 0$ and all $\theta \in \mathbb{R}$, we let $\mathcal{L}_{\alpha,\epsilon}(\theta) \in (0, \infty)$ be the smallest constant $L \in (0, \infty)$ such that

$$|\nabla F(\theta) - \nabla F(\theta')| \leq L \|\theta - \theta'\|^{1+\alpha}, \quad \forall \theta' \in B_{r_{\alpha,\epsilon,\theta}}(\theta),$$

where $r_{\alpha,\epsilon,\theta} = (G_\alpha(\theta) \vee \epsilon)^{\frac{1}{1+\alpha}}$ and where $B_r(\theta)$ denote the closed ball centred at $\theta \in \mathbb{R}$ and with radius $r > 0$.

Then, to show that Assumption 5 of [16] does not hold it is enough to show that

$$(8) \quad \liminf_{\theta \rightarrow \infty} \frac{\mathcal{L}_{\alpha,\epsilon}(\theta) G_\alpha(\theta)}{F(\theta) + |\nabla F(\theta)|^2} = \infty, \quad \forall \epsilon > 0, \quad \forall \alpha \in (0, 1].$$

To show (8) let $\alpha \in (0, 1]$ and $\epsilon > 0$, and we first compute a lower bound for $G_\alpha(\theta)$ and for $\mathcal{L}_{\alpha,\epsilon}(\theta)$. Using Hölder inequality, for all $\theta \in \mathbb{R}$ we have

$$\mathbb{E}[|\nabla f(\theta, Z)|] \leq \mathbb{E}[|\nabla f(\theta, Z)|^{1+\alpha}]^{\frac{1}{1+\alpha}},$$

where

$$\begin{aligned} \mathbb{E}[|\nabla f(\theta, Z)|] &= \mathbb{E}[|X \exp(\theta X) - XY|] \geq \mathbb{E}[X \exp(\theta X)] - \mathbb{E}[XY] \\ &= \mathbb{E}[X \exp(\theta X)] - 1 \\ &= \exp(\exp(\theta) + \theta - 1) \end{aligned}$$

and thus

$$(9) \quad G_\alpha(\theta) \geq \left(\exp(\exp(\theta) + \theta - 1) - 1 \right)^{1+\alpha}, \quad \forall \theta \geq 1.$$

To proceed further let $\theta \in \mathbb{R}$. To obtain a lower bound for $\mathcal{L}_{\alpha,\epsilon}(\theta)$ let $r > \log 2$ and, using the fact that

$$\exp(\theta') \geq \exp(\theta) + (\theta' - \theta) \exp(\theta), \quad \forall \theta' \in \mathbb{R},$$

remark that

$$\exp(\exp(\theta) - \exp(\theta') + \theta - \theta') \leq \frac{1}{2}, \quad \forall \theta' \in (\theta + r, \infty).$$

Therefore, for all $\theta' \in (\theta + r, \infty)$ we have, using (6),

$$\begin{aligned} |\nabla F(\theta) - \nabla F(\theta')| &= \left| \exp(\exp(\theta) + \theta - 1) - \exp(\exp(\theta') + \theta' - 1) \right| \\ &= \exp(\exp(\theta') + \theta' - 1) \left(1 - \exp(\exp(\theta) - \exp(\theta') + \theta - \theta') \right) \\ &\geq \frac{1}{2} \exp(\exp(\theta') + \theta' - 1) \\ &= \frac{\exp(\exp(\theta') + \theta' - 1)}{2|\theta' - \theta|^\alpha} |\theta' - \theta|^\alpha \end{aligned}$$

and thus

$$(10) \quad |\nabla F(\theta) - \nabla F(\theta')| \geq \frac{\exp(\exp(\theta + r) + \theta + r - 1)}{2(2r)^\alpha} |\theta' - \theta|^\alpha, \quad \forall \theta' \in (B_{2r}(\theta) \setminus B_r(\theta)) \cap (\theta, \infty).$$

This shows that if $D_{r,\alpha}(\theta) \in (0, \infty)$ is such that

$$|\nabla F(\theta) - \nabla F(\theta')| \leq D_{r,\alpha}(\theta) |\theta - \theta'|^\alpha, \quad \forall \theta' \in B_{2r}(\theta),$$

then

$$D_{r,\alpha}(\theta) \geq K_{r,\alpha}(\theta) := \frac{\exp(\exp(\theta + r) + \theta + r - 1)}{2(2r)^\alpha}.$$

To proceed further let $c_{\alpha,\epsilon} \in (1, \infty)$ be such that

$$\exp(\exp(\theta) + \theta - 1) - 1 \geq R(\theta) := \frac{1}{2} \exp(\exp(\theta) + \theta - 1) \geq \epsilon^{\frac{1}{1+\alpha}}, \quad \forall \theta \geq c_{\alpha,\epsilon}$$

and note that, by (9), we have $r_{\alpha,\epsilon,\theta} \geq R(\theta)$ for all $\theta \geq c_{\alpha,\epsilon}$.

We now let $c'_{\alpha,\epsilon} \in \mathbb{R}$ be such that $c'_{\alpha,\epsilon} > c_{\alpha,\epsilon}$ and such that $R(\theta) > 2$ for all $\theta \geq c'_{\alpha,\epsilon}$. Then, $r_{\alpha,\epsilon,\theta} > 2 > \log 2$ for all $\theta \geq c'_{\alpha,\epsilon}$ and thus, using the above computations,

$$(11) \quad \mathcal{L}_{\alpha,\epsilon}(\theta) \geq K_{\frac{1}{2}r_{\alpha,\epsilon,\theta},\alpha}(\theta), \quad \forall \theta \geq c'_{\alpha,\epsilon}.$$

Therefore, using (9) and (11), and noting that for all $\theta > 0$ we have $K_{r',\alpha}(\theta) \geq K_{r,\alpha}(\theta)$ for all $r' \geq r \geq 1$, it follows that for all $\theta \geq c'_{\alpha,\epsilon}$ we have

$$\begin{aligned} \mathcal{L}_{\alpha,\epsilon}(\theta)G_\alpha(\theta) &\geq R(\theta)^{1+\alpha}K_{\frac{1}{2}r_{\alpha,\epsilon,\theta},\alpha} \\ &\geq R(\theta)^{1+\alpha}K_{\frac{1}{2}R(\theta),\alpha} \\ &= R(\theta)^{1+\alpha} \frac{\exp(\exp(\theta + R(\theta)/2) + \theta + R(\theta)/2 - 1)}{2R(\theta)^\alpha} \\ &= \frac{R(\theta)}{2} \exp(\exp(\theta + R(\theta)/2) + \theta + R(\theta)/2 - 1) \end{aligned}$$

and thus

$$\liminf_{\theta \rightarrow \infty} \frac{\mathcal{L}_{\alpha,\epsilon}(\theta)G_\alpha(\theta)}{F(\theta) + |\nabla F(\theta)|^2} = \infty.$$

Since $\alpha \in (0, 1]$ and $\epsilon > 0$ are arbitrary, (8) follows and the proof of the proposition is complete. $\square$

A.2. Proofs of the results of Section 1.

A.2.1. Proof of Proposition 3.

Proof of Proposition 3. From (6) we observe that $\nabla^2 F(\theta) > 0$ for all $\theta \in \mathbb{R}$ and thus the function F satisfies A1. Remark also that, from (6), the unique minimizer of F is $\theta_\star = 0$.

The function $\theta \mapsto f(\theta, z)$ is twice continuously differentiable on $\mathbb{R}$ for all $z \in \mathbb{R}^2$ where, for all $\theta \in \mathbb{R}$ and $z \in \mathbb{R}^2$,

$$(12) \quad \nabla f(\theta, z) = -z_1 z_2 + z_1 \exp(\theta z_1), \quad \nabla^2 f(\theta, z) = z_1^2 \exp(\theta z_1).$$

Using (12) and the fact that $\theta_\star = 0$, we obtain

$$(13) \quad \mathbb{E}[(\nabla f(\theta_\star, Z))^2] = \mathbb{E}[(-XY + X \exp(\theta_\star X))^2] = \mathbb{E}[X^2(1 - Y)^2] = 2$$

and

$$(14) \quad \mathbb{E}[|\nabla^2 f(\theta_\star, Z)|] = \mathbb{E}[X^2 \exp(\theta_\star X)] = 2.$$

We now let $\Theta \subset \mathbb{R}$ be a non-empty compact set, $C_\Theta \in (0, \infty)$ be such that $|\theta| \leq C_\Theta$ and such that $|\theta - \theta'| \leq C_\Theta$ for all $\theta, \theta' \in \Theta$, and we let $M_\Theta : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$M_\Theta(z) = |z_1^3| \exp(2C_\Theta |z_1|), \quad z \in \mathbb{R}^2.$$

Then, using the fact that $|e^x - 1| \leq |x|e^{|x|}$ for all $x \in \mathbb{R}$, for all $(\theta, \theta') \in \Theta^2$ and $z \in \mathbb{R}^2$ we have

$$\begin{aligned} |\nabla^2 f(\theta, z) - \nabla^2 f(\theta', z)| &= z_1^2 |\exp(\theta z_1) - \exp(\theta' z_1)| \\ &= z_1^2 \exp(\theta z_1) |1 - \exp((\theta' - \theta) z_1)| \\ &\leq |z_1^3| \exp(\theta z_1) |\theta' - \theta| \exp(|\theta' - \theta| |z_1|) \\ &\leq M_\Theta(z) |\theta' - \theta|, \end{aligned}$$

where, using Cauchy-Schwartz inequality,

$$\begin{aligned} \mathbb{E}[M_\Theta(Z)] &= \mathbb{E}[X^3 \exp(2C_\Theta X)] \leq \left(\mathbb{E}[X^6] \mathbb{E}[\exp(4C_\Theta X)] \right)^{1/2} \\ &= \mathbb{E}[X^6]^{1/2} \exp\left(\exp(4C_\Theta) - 1\right)^{1/2} \\ &< \infty. \end{aligned}$$

Together with (13)–(14) this shows that A2 holds and the proof of Proposition 3 is complete. $\square$

A.2.2. Proof of Lemma 1.

Proof of Lemma 1. Let $\Theta \subset \mathbb{R}^d$ be a (non-empty) closed ball containing $\theta_\star$ and let $(\theta, \theta') \in \Theta^2$.

Then, under A2, by Taylor's theorem there exists a Θ -valued random variable $\tilde{\theta}$ such that, $\mathbb{P}$ -a.s.,

$$\begin{aligned} \|\nabla f(\theta, Z) - \nabla f(\theta', Z)\| &= \|\nabla^2 f(\tilde{\theta}, Z)(\theta - \theta')\| \\ &\leq \|\theta - \theta'\| \|\nabla^2 f(\tilde{\theta}, Z)\| \\ (15) \quad &\leq \|\theta - \theta'\| \left(\|\nabla^2 f(\tilde{\theta}, Z) - \nabla^2 f(\theta_\star, Z)\| + \|\nabla^2 f(\theta_\star, Z)\| \right) \\ &\leq \|\theta - \theta'\| \left(\|\tilde{\theta} - \theta_\star\| M_\Theta(Z) + \|\nabla^2 f(\theta_\star, Z)\| \right) \\ &\leq \|\theta - \theta'\| M_\Theta^{(1)}(Z), \end{aligned}$$

where

$$M_\Theta^{(1)}(z) = \sup_{\theta_1, \theta_2 \in \Theta} \|\theta_1 - \theta_2\| M_\Theta(z) + \|\nabla^2 f(\theta_\star, z)\|, \quad \forall z \in \mathbb{Z}.$$

Under A2 we have both $\mathbb{E}[M_\Theta(Z)] < \infty$ and $\mathbb{E}[\|\nabla^2 f(\theta_\star, Z)\|] < \infty$, and thus $\mathbb{E}[M_\Theta^{(1)}(Z)] < \infty$.

Similarly, under A2, by Taylor's theorem there exists a Θ -valued random variable $\tilde{\theta}$ such that, $\mathbb{P}$ -a.s.,

$$\begin{aligned} |f(\theta, Z) - f(\theta', Z)| &= |(\theta - \theta')^\top \nabla f(\tilde{\theta}, Z)| \\ &\leq \|\theta - \theta'\| \|\nabla f(\tilde{\theta}, Z)\| \\ &\leq \|\theta - \theta'\| \left(\|\nabla f(\tilde{\theta}, Z) - \nabla f(\theta_\star, Z)\| + \|\nabla f(\theta_\star, Z)\| \right) \\ &\leq \|\theta - \theta'\| \left(\|\tilde{\theta} - \theta_\star\| M_\Theta^{(1)}(Z) + \|\nabla f(\theta_\star, Z)\| \right) \\ &\leq \|\theta - \theta'\| M_\Theta^{(2)}(Z), \end{aligned}$$

where the third inequality uses (15) and where

$$M_{\Theta}^{(2)}(z) = \sup_{\theta_1, \theta_2 \in \Theta} \|\theta_1 - \theta_2\| M_{\Theta}^{(1)}(z) + \|\nabla f(\theta_*, z)\|, \quad \forall z \in \mathcal{Z}.$$

Since, as shown above, $\mathbb{E}[M_{\Theta}^{(1)}(Z)] < \infty$ while, by assumption, $\mathbb{E}[\|\nabla f(\theta_*, Z)\|] < \infty$, it follows that $\mathbb{E}[M_{\Theta}^{(2)}(Z)] < \infty$. The result of the lemma follows. $\square$

A.2.3. Proof of Lemma 2.

Proof of Lemma 2. Let θ_* be as in A2 and Θ be a (non-empty) closed ball containing θ_* . Next, let $\theta \in \Theta$, $M'_{\Theta}(\cdot)$ be as in Lemma 1 and

$$G(z) = \sup_{\theta_1, \theta_2 \in \Theta} \|\theta_1 - \theta_2\| M'_{\Theta}(z) + \|\nabla f(\theta_*, z)\| + \|\nabla^2 f(\theta_*, z)\|, \quad \forall z \in \mathcal{Z}.$$

Then, for all $k \in \{1, 2\}$ we $\mathbb{P}$ -a.s. have, by Lemma 1,

$$\begin{aligned} \sup_{\theta \in \Theta} \|\nabla^k f(\theta, Z)\| &\leq \sup_{\theta \in \Theta} \|\nabla^k f(\theta, Z) - \nabla^k f(\theta_*, Z)\| + \|\nabla^k f(\theta_*, Z)\| \\ &\leq \sup_{\theta_1, \theta_2 \in \Theta} \|\theta_1 - \theta_2\| M'_{\Theta}(Z) + \|\nabla^k f(\theta_*, Z)\| \\ &\leq G(Z). \end{aligned}$$

Since by Lemma 1 and under A2 we have $\mathbb{E}[G(Z)] < \infty$, it follows from the dominated convergence theorem that, for all $k \in \{1, 2\}$, we have $\nabla^k F(\theta) = \mathbb{E}[\nabla^k f(\theta, Z)]$ for all $\theta \in \Theta$ and thus, since Θ is arbitrary, $\nabla^k F(\theta) = \mathbb{E}[\nabla^k f(\theta, Z)]$ for all $\theta \in \mathbb{R}^d$.

Since F is twice differentiable on $\mathbb{R}^d$, the mapping $\nabla F : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is continuous and thus to conclude the proof of the lemma it remains to show that the mapping $\nabla^2 F : \mathbb{R}^d \rightarrow \mathbb{R}^{d \times d}$ is continuous as well. To this aim, let Θ be a (non-empty) closed ball containing θ_* and let $\theta, \theta' \in \Theta$. Then,

$$\begin{aligned} \|\nabla^2 F(\theta) - \nabla^2 F(\theta')\| &= \|\mathbb{E}[\nabla^2 f(\theta, Z)] - \mathbb{E}[\nabla^2 f(\theta', Z)]\| \\ &\leq \mathbb{E}[\|\nabla^2 f(\theta, Z) - \nabla^2 f(\theta', Z)\|] \\ &\leq \|\theta - \theta'\| \mathbb{E}[M_{\Theta}(Z)], \end{aligned}$$

where the first inequality holds by Jensen's inequality and the second one holds under A2. Since $\mathbb{E}[M_{\Theta}(Z)] < \infty$, this shows that the mapping $\nabla^2 F : \mathbb{R}^d \rightarrow \mathbb{R}^{d \times d}$ is locally Lipschitz and thus continuous. The result of the lemma follows. $\square$

A.2.4. Proof of Lemma 3.

Proof of Lemma 3. Let $k \in \{0, 1, 2\}$ and note that, using Lemma 1, we have $\mathbb{E}[\|\nabla^k f(\theta, Z)\|] < \infty$ for all $\theta \in \mathbb{R}^d$ under A2. Therefore, by the law of large numbers, for all $\theta \in \mathbb{R}^d$ we have

$$\lim_{n \rightarrow \infty} \left\| \nabla^k F_n(\theta) - \mathbb{E}[\nabla^k f(\theta, Z)] \right\| = 0, \quad \mathbb{P} - a.s.$$

To proceed further let $\Theta \subset \mathbb{R}^d$ be a (non-empty) compact set. Then, by Lemma 1, there exists a measurable function $M'_{\Theta}(\cdot)$ such that $\mathbb{E}[M'_{\Theta}(Z)] < \infty$ and such that

$$\|\nabla^k f(\theta, Z) - \nabla^k f(\theta', Z)\| \leq M'_{\Theta}(Z) \|\theta - \theta'\|, \quad \forall \theta, \theta' \in \Theta, \quad \mathbb{P} - a.s.$$

Therefore, $\mathbb{P}$ -a.s., for all $\theta, \theta' \in \mathbb{R}^d$ and $n \geq 1$ we have

$$\begin{aligned}
 \|\nabla^k F_n(\theta) - \nabla^k F_n(\theta')\| &= \frac{1}{n} \left\| \sum_{i=1}^n \left(\nabla^k f(\theta, Z_i) - \nabla^k f(\theta', Z_i) \right) \right\| \\
 (16) \qquad \qquad \qquad &\leq \frac{1}{n} \sum_{i=1}^n \|\nabla^k f(\theta, Z_i) - \nabla^k f(\theta', Z_i)\| \\
 &\leq \|\theta - \theta'\| \frac{1}{n} \sum_{i=1}^n M'_\Theta(Z_i).
 \end{aligned}$$

By the law of large numbers, $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n M'_\Theta(Z_i) = \mathbb{E}[M'_\Theta(Z)] < \infty$, $\mathbb{P}$ -a.s., and thus

$$(17) \qquad \qquad \qquad \frac{1}{n} \sum_{i=1}^n M'_\Theta(Z) = \mathcal{O}(1), \quad \mathbb{P} - a.s.$$

Using (16)–(17), as well as the fact that Θ is compact and the fact that, by Lemma 2, the mapping $\nabla^k F$ is uniformly continuous on the compact set Θ , it is readily checked that

$$\lim_{n \rightarrow \infty} \sup_{\theta \in \Theta} \left\| \nabla^k F_n(\theta) - \mathbb{E}[\nabla^k f(\theta, Z)] \right\| = 0.$$

The proof of the lemma is complete. $\square$

A.2.5. Proof of Lemma 4.

Proof of Lemma 4. To simplify the notation in what follows for all $\theta \in \mathbb{R}^d$ we let $H(\theta) = \mathbb{E}[\nabla^2 f(\theta, Z)]$ and $H_n(\theta) = \nabla^2 F_n(\theta)$ for all $n \geq 1$.

Let Θ be a (non-empty) compact set, $\theta \in \Theta$ and $n \geq 1$. Note that the matrices $H_n(\theta)$ and $H(\theta)$ are symmetric and thus, by the min-max theorem,

$$\begin{aligned}
 \lambda_{\min}(H_n(\theta)) &= \min \left\{ x^\top H_n(\theta) x : \|x\| = 1 \right\}, \\
 (18) \qquad \lambda_{\min}(H(\theta)) &= \min \left\{ x^\top H(\theta) x : \|x\| = 1 \right\}.
 \end{aligned}$$

Since for all $x \in \mathbb{R}^d$ such that $\|x\| = 1$ we have

$$x^\top H_n(\theta) x = x^\top H(\theta) x + x^\top (H_n(\theta) - H(\theta)) x \geq \lambda_{\min}(H(\theta)) - \|H_n(\theta) - H(\theta)\|,$$

it follows from Lemma 3 that

$$\begin{aligned}
 \liminf_{n \rightarrow \infty} \inf_{\theta \in \Theta} \lambda_{\min}(H_n(\theta)) &\geq \inf_{\theta \in \Theta} \lambda_{\min}(H(\theta)) - \limsup_{n \rightarrow \infty} \sup_{\theta \in \Theta} \|H_n(\theta) - H(\theta)\| \\
 &= \inf_{\theta \in \Theta} \lambda_{\min}(H(\theta)).
 \end{aligned}$$

By Lemma 2, the mapping $H : \mathbb{R}^d \rightarrow \mathbb{R}^{d \times d}$ is continuous and thus, using (18) and the maximum theorem, it follows that the mapping $\theta \mapsto \lambda_{\min}(H(\theta))$ is continuous. Therefore, since Θ is compact, there exists a $\theta_\Theta \in \Theta$ such that

$$\inf_{\theta \in \Theta} \lambda_{\min}(H(\theta)) = \lambda_{\min}(H(\theta_\Theta))$$

which concludes the proof of the first part of the lemma.

Using the fact, by the min-max theorem,

$$\begin{aligned}\lambda_{\max}(H_n(\theta)) &= \max \left\{ x^\top H_n(\theta) x : \|x\| = 1 \right\}, \\ \lambda_{\max}(H(\theta)) &= \max \left\{ x^\top H(\theta) x : \|x\| = 1 \right\},\end{aligned}$$

the second part of the lemma is proved in a similar way and its proof is therefore omitted to save space. $\square$

A.2.6. Proof of Lemma 5.

Proof of Lemma 5. Let Θ be a compact and convex set such that $\theta_\star \in \mathring{\Theta}$ and remark that, by Lemma 2, we have $\mathbb{E}[\nabla f(\theta_\star, Z)] = 0$. Hence, since by assumption we have $\mathbb{E}[\|\nabla f(\theta_\star, Z)\|^{1+\alpha}] < \infty$, it follows from Lemma 3 in [4] that there exists a constant $C^\star < \infty$ such that

$$(19) \quad \mathbb{P}(\|\nabla F_n(\theta_\star)\| \geq M) \leq \frac{C^\star}{n^\alpha M^{1+\alpha}}, \quad \forall M > 0, \quad \forall n \geq 1.$$

Remark also that, since $\sup_{\theta \in \Theta} \|F_n(\theta) - F(\theta)\| = o_{\mathbb{P}}(1)$ by Lemma 3 while, under A1, there exists a constant $c_\Theta > 0$ such that $F(\theta) - F(\theta_\star) \geq c_\Theta \|\theta - \theta_\star\|$ for all $\theta \in \Theta$, it follows from Theorem 5.7 in [19, page 45] that

$$(20) \quad \|\theta_\star - \hat{\theta}_{\Theta, n}\| = o_{\mathbb{P}}(1).$$

We now let

$$\lambda_\Theta = \frac{1}{2} \inf_{\theta \in \Theta} \lambda_{\min}(\mathbb{E}[\nabla^2 f(\theta, Z)])$$

and, to simplify the notation in what follows, for all $\theta \in \mathbb{R}^d$ and $n \geq 1$ we let $H_n(\theta) = \nabla^2 F_n(\theta)$. Finally, for all $\epsilon > 0$ and $n \geq 1$ we let

$$\begin{aligned}\Omega_{\epsilon, n} &= \left\{ \omega \in \Omega : \hat{\theta}_{\Theta, n}^\omega \in \mathring{\Theta}, \|\nabla F_n^\omega(\theta_\star)\| \leq n^{-\frac{\alpha}{1+\alpha}} (C^\star/\epsilon)^{\frac{1}{1+\alpha}}, \right. \\ &\quad \left. \inf_{\theta \in \Theta} \lambda_{\min}(H_n^\omega(\theta)) \geq \lambda_\Theta \right\}.\end{aligned}$$

Remark that, by (19)–(20) and Lemma 4, and since $\theta_\star \in \mathring{\Theta}$, it follows that we have $\liminf_{n \rightarrow \infty} \mathbb{P}(\Omega_{\epsilon, n}) \geq 1 - \epsilon$ for all $\epsilon \in (0, 1)$. Remark also that $\lambda_\Theta > 0$ under A1 and by Lemma 2, and thus for all $n \geq 1$ and $\epsilon > 0$ the function F_n^ω is strictly convex on Θ for all $\omega \in \Omega_{\epsilon, n}$.

Let $n \geq 1$, $\epsilon \in (0, 1)$ and $\omega \in \Omega_{\epsilon, n}$, and note that $\nabla F_n^\omega(\hat{\theta}_{\Theta, n}^\omega) = 0$ since F_n^ω is strictly convex on Θ and $\hat{\theta}_{\Theta, n}^\omega \in \mathring{\Theta}$. Therefore, since $\theta_\star \in \mathring{\Theta}$, it follows from Taylor's theorem that there exists a $\check{\theta}_{n_B}^\omega \in \mathring{\Theta}$ such that

$$(21) \quad \nabla F_n^\omega(\theta_\star) = H_n^\omega(\check{\theta}_n^\omega)(\theta_\star - \hat{\theta}_{\Theta, n}^\omega) \Leftrightarrow \theta_\star - \hat{\theta}_{\Theta, n}^\omega = (H_n^\omega(\check{\theta}_n^\omega))^{-1} \nabla F_n^\omega(\theta_\star).$$

Remark that the matrix $H_n^\omega(\check{\theta}_n^\omega)$ is indeed invertible since all its eigenvalues are bounded below by λ_Θ , where $\lambda_\Theta > 0$ under A1 and by Lemma 2.

Using (21), we obtain that

$$\|\theta_\star - \hat{\theta}_{\Theta, n}^\omega\| \leq \|H_n^\omega(\check{\theta}_n^\omega)\|^{-1} \|\nabla F_n^\omega(\theta_\star)\| \leq \left(\frac{C^\star}{\epsilon}\right)^{\frac{1}{1+\alpha}} \lambda_\Theta^{-1} n^{-\frac{\alpha}{1+\alpha}}$$

and the result of the lemma follows. $\square$

A.2.7. *Proof of Proposition 4.*

Proof of Proposition 4. We consider the optimization problem (1) with $d = 1$, with $Z = \mathbb{R}$, with $f : \mathbb{R} \times Z \rightarrow \mathbb{R}$ defined by $f(\theta, z) = \theta^2 - 2\theta z$, $(\theta, z) \in \mathbb{R}^2$ and with $Z \sim P$ for some probability distribution P . Then, it is direct to check that for any distribution P such that $\mathbb{E}[|Z|^{1+\alpha'}] < \infty$ Assumptions A1–A2 hold with $\alpha = \alpha'$, and that we have $\theta_* = \mathbb{E}[Z]$. For all $B \in \mathbb{N}$ let $\hat{\theta}_B$ be an estimator of θ_* based on $k_B \leq B$ distinct elements of $(Z_i)_{i \geq 1}$. Then, from the proof of Theorem 3.1 in [5], it is direct to see that there exists a distribution P of Z such that $\mathbb{E}[|Z|^{1+\alpha'}] < \infty$ and such that, for some constants $\delta \in (0, 1)$ and $C \in (0, \infty)$, we have

$$\limsup_{B \rightarrow \infty} \mathbb{P}\left(|\hat{\theta}_B - \mathbb{E}[Z]| > C k_B^{-\frac{\alpha}{1+\alpha}}\right) \geq \delta.$$

The result of the proposition follows upon noting that $k_B \leq B$ for all $B \in \mathbb{N}$. $\square$

A.2.8. *Proof of Proposition 5.*

Proof of Proposition 5. Since g is convex on the convex set K and $x_t \in K$ for all $t \geq 0$, it follows that

$$g(x_*) \geq g(x_t) + \nabla g(x_t)^\top (x_* - x_t), \quad \forall t \geq 0$$

and thus

$$(22) \quad g(x_t) \leq g(x_*) - \nabla g(x_t)^\top (x_* - x_t) = g(x_*) + \nabla g(x_t)^\top (x_t - x_*), \quad \forall t \geq 0.$$

Remark that for all $t \geq 0$ we have

$$(23) \quad \|x_{t+1} - x_*\|^2 = \|x_t - x_*\|^2 + v_t^2 \|\nabla g(x_t)\|^2 - 2v_t \nabla g(x_t)^\top (x_t - x_*)$$

and let $t \geq 0$. Then,

$$\begin{aligned} & g(x_{t+1}) - g(x_*) \\ & \leq g(x_t) - g(x_*) - \frac{v_t}{2} \|g(x_t)\|^2 \\ & \leq \nabla g(x_t)^\top (x_t - x_*) - \frac{v_t}{2} \|g(x_t)\|^2 \\ (24) \quad & = \frac{1}{2v_t} \left(2v_t \nabla g(x_t)^\top (x_t - x_*) - v_t^2 \|g(x_t)\|^2 \right) \\ & = \frac{1}{2v_t} \left(2v_t \nabla g(x_t)^\top (x_t - x_*) - v_t^2 \|g(x_t)\|^2 - \|x_t - x_*\|^2 + \|x_t - x_*\|^2 \right) \\ & = \frac{1}{2v_t} (\|x_t - x_*\|^2 - \|x_{t+1} - x_*\|^2), \end{aligned}$$

where the first inequality uses (2), the second inequality uses (22) and the third equality uses (23).

In addition, since $g(x_{t+1}) - g(x_*) \geq 0$ for all $t \geq 0$, it follows from (24) that

$$(25) \quad \|x_t - x_*\|^2 - \|x_{t+1} - x_*\|^2 \geq 0, \quad \forall t \geq 1$$

and thus, since for some constant $c > 0$ we have $\inf_{t \geq 1} v_t \geq c > 0$ by assumption, it follows from (24)–(25) that

$$g(x_{t+1}) - g(x_*) \leq \frac{1}{2c} (\|x_t - x_*\|^2 - \|x_{t+1} - x_*\|^2), \quad \forall t \geq 0.$$

To conclude the proof let $T \in \mathbb{N}$. Then, using this latter result, we have

$$\begin{aligned} \sum_{t=1}^T (g(x_t) - g(x_*)) &\leq \frac{1}{2c} \sum_{t=1}^T \left(\|x_{t-1} - x_*\|^2 - \|x_t - x_*\|^2 \right) \\ &= \frac{1}{2c} (\|x_0 - x_*\|^2 - \|x_T - x_*\|^2) \\ &\leq \frac{\|x_0 - x_*\|^2}{2c} \end{aligned}$$

and thus, since by (2) we have $g(x_t) - g(x_*) \leq g(x_{t-1}) - g(x_*)$ for all $t \geq 1$, it follows that

$$g(x_T) - g(x_*) \leq \frac{1}{T} \sum_{t=1}^T (g(x_t) - g(x_*)) \leq \frac{\|x_0 - x_*\|^2}{2cT}$$

and the proof of the proposition is complete. $\square$

A.2.9. Proof of Proposition 6.

Proof of Proposition 6. Let K and $\tilde{K}$ be as in the statement of the proposition, and remark that since g is convex and L_K -smooth on the convex set K we have, by the descent lemma,

$$(26) \quad g(y) \leq g(x) + \nabla g(x)^\top (y - x) + \frac{L_K}{2} \|y - x\|^2, \quad \forall (x, y) \in K^2.$$

Remark also that $x - v\nabla g(x) \in K$ for all $x \in \tilde{K}$ and all $v \in (0, 1]$, and let $x \in \tilde{K} \subset K$ and $v \in (0, 1]$. Then, since by (26) we have

$$g(x - v\nabla g(x)) \leq g(x) - v\left(1 - \frac{L_K}{2}v\right)\|\nabla g(x)\|^2,$$

where

$$v\left(1 - \frac{L_K}{2}v\right) \geq \frac{v}{2} \Leftrightarrow v \leq \frac{1}{L_K},$$

it follows that if $v = \beta^{k_x}$ with

$$k_x = \min \left\{ k \in \mathbb{N}_0 : g(x - \beta^k \|\nabla g(x)\|) \leq g(x) - \frac{\beta^k}{2} \|\nabla g(x)\|^2 \right\},$$

then $v \geq \beta/L_K$. Since $x_t \in \tilde{K}$ for all $t \geq 0$, this shows that $v_t \geq \beta/L_K$ for all $t \geq 1$ and the proof of the proposition is complete. $\square$

A.3. Preliminary results for proving Theorems 1 and 2.

A.3.1. A technical lemma.

Lemma 6. Let $\alpha' \in (0, 1]$, $\delta \in (0, 2\alpha'/(1+3\alpha'))$ and, for all $j \in \mathbb{N}$, let $\gamma_j = 1 - \delta^j$. Next, let $w_{\alpha'} = (1 + \alpha')/(1 + 3\alpha')$ and

$$\gamma'_j = w_{\alpha'} + (1 - w_{\alpha'})\gamma_{j-1}, \quad \forall j \in \mathbb{N} \setminus \{1\}.$$

Then, there exists a constant $\pi \in (0, 1)$ such that

$$\gamma_j = \pi\gamma'_j + (1 - \pi)\gamma_{j-1}, \quad \forall j \in \mathbb{N} \setminus \{1\}.$$

In addition, if for some constants $(\kappa, \tau) \in (0, \infty)^2$ and all $j \in \mathbb{N}$ and $B \in \mathbb{N}$ we let $n_{j,B} = \lceil \kappa B^{\gamma_j} \rceil$, $\tau_{j,B} = \tau B^{-\frac{\alpha'}{1+\alpha'}\gamma_j}$ and $r_{j,B} = B^{-\frac{\alpha'}{1+\alpha'}\gamma_j}$, then

$$\lim_{B \rightarrow \infty} \frac{n_{j,B} r_{j-1,B}^2}{\tau_{j,B}^2 B} = \lim_{B \rightarrow \infty} \frac{n_{j,B}}{B} = 0, \quad \forall j \in \mathbb{N}$$

with the convention that $r_{j-1,B} = 1$ for all $B \in \mathbb{N}$ when $j = 1$.

Proof of Lemma 6. We show that the first part of the lemma holds for

$$\pi = (1 - \delta) \frac{1 + 3\alpha'}{1 + \alpha'}.$$

Remark that the conditions on δ ensure that $\pi \in (0, 1)$ and that, by definition, π is such that

$$(27) \quad \frac{2\alpha' + (1 - \pi)(1 + \alpha')}{1 + 3\alpha'} = \delta.$$

Then, the first part of the lemma holds if and only if for all $j \geq 2$ we have

$$\begin{aligned} \gamma_j &= \pi w_{\alpha'} + \pi(1 - w_{\alpha'})\gamma_{j-1} + (1 - \pi)\gamma_{j-1} \\ &= \pi w_{\alpha'} + (1 - \pi w_{\alpha'})\gamma_{j-1} \\ &= \pi w_{\alpha'} \sum_{i=0}^{j-2} (1 - \pi w_{\alpha'})^i + (1 - \pi w_{\alpha'})^{j-1} \gamma_1 \\ (28) \quad &= 1 - (1 - \pi w_{\alpha'})^{j-1} (1 - \gamma_1) \\ &= 1 - \left(\frac{2\alpha' + (1 - \pi)(1 + \alpha')}{1 + 3\alpha'} \right)^{j-1} (1 - \gamma_1) \\ &= 1 - \delta^{j-1} (1 - \gamma_1) \\ &= 1 - \delta^j, \end{aligned}$$

where the penultimate equality uses (27) and the last one the fact that $\gamma_1 = 1 - \delta$. This shows the first part of the lemma.

To show the second part of the lemma note that, by the first part of the lemma, we have

$$(29) \quad \gamma_j < \gamma_{j+1} < \gamma'_{j+1}, \quad \forall j \in \mathbb{N}$$

and to simplify the notation in the following let $c_{\alpha'} = \alpha'/(1 + \alpha')$.

Firstly, the conditions on δ ensuring that $\gamma_1(1 + 2c_{\alpha'}) - 1 < 0$, it follows that

$$(30) \quad \lim_{B \rightarrow \infty} \frac{n_{1,B}}{\tau_{1,B}^2 B} = 0.$$

Next, using the first part of the lemma, we remark that for all $j \in \mathbb{N} \setminus \{1\}$ we have

$$\begin{aligned} &(1 + 2c_{\alpha'})\gamma_j - 2c_{\alpha'}\gamma_{j-1} - 1 \\ &= (1 + 2c_{\alpha'})\gamma'_j - 2c_{\alpha'}\gamma_{j-1} - 1 + (1 + 2c_{\alpha'})(1 - \pi)(\gamma_{j-1} - \gamma'_j) \\ &= (1 + 2c_{\alpha'})(1 - \pi)(\gamma_{j-1} - \gamma'_j) \\ &< 0, \end{aligned}$$

where the last inequality uses (29), and thus

$$(31) \quad \lim_{B \rightarrow \infty} \frac{n_{j+1,B} r_{j,B}^2}{\tau_{j+1,B}^2 B} = 0, \quad \forall j \in \mathbb{N}.$$

By combining (30) and (31) we obtain that

$$(32) \quad \lim_{B \rightarrow \infty} \frac{n_{j,B} r_{j-1,B}^2}{\tau_{j,B}^2 B} = 0, \quad \forall j \in \mathbb{N}.$$

Finally, since $\gamma_j < 1$ for all $j \in \mathbb{N}$, we have

$$(33) \quad \lim_{B \rightarrow \infty} \frac{n_{j,B}}{B} = 0, \quad \forall j \in \mathbb{N}$$

and the second part of the lemma follows from (32)–(33). The proof is complete. $\square$

A.3.2. A first key property of Algorithm 3.

Lemma 7. Assume that A1–A2 hold. Let $x \in \mathbb{R}^d$, $\beta \in (0, 1)$, $(n_B)_{B \geq 1}$ be a sequence in $\mathbb{N}$, $(\tau_B)_{B \geq 1}$ be a sequence in $[0, \infty)$ and $(\tilde{B}_B)_{B \geq 1}$ be a sequence of $\mathbb{N}_0$ -valued random variables. In addition, for all $B \in \mathbb{N}$ and $\theta_0 \in \mathbb{R}^d$, let

$$(\hat{\theta}_{\theta_0,B}, \tilde{B}_{\theta_0}) = \text{Algorithm 3}(\theta_0, n_B, \tilde{B}_B, \tau_B, \beta)$$

and denote by $\{\theta_{\theta_0,B,t}\}_{t=0}^{T_{\theta_0,B}}$ the sequence in $\mathbb{R}^d$ generated by Algorithm 3 to compute $\hat{\theta}_{\theta_0,B}$ from θ_0 and (which is therefore such that $\theta_{\theta_0,B,0} = \theta_0$ and such that $\theta_{\theta_0,B,T_{\theta_0,B}} = \hat{\theta}_{\theta_0,B}$). Finally, for all $j \in \mathbb{N}_0$, let $\Theta_{j,x} = \{\theta \in \mathbb{R}^d : F(\theta) \leq F(x) + j\}$. Then, for all $j \in \mathbb{N}_0$, the set $\Theta_{j+1,x}$ is a (non-empty) compact and convex set such that $\theta_\star \in \dot{\Theta}_{j+1,x}$, and there exists a sequence $(\Omega_{j,B}^{(1)})_{B \geq 1}$ in $\mathcal{F}$ such that $\lim_{B \rightarrow \infty} \mathbb{P}(\Omega_{j,B}^{(1)}) = 1$ and such that

$$\theta_{\theta_0,B,t}^\omega \in \Theta_{j+1,x}, \quad \forall t \in \{0, \dots, T_{\theta_0,B}^\omega\}, \quad \forall \omega \in \Omega_{j,B}^{(1)}, \quad \forall \theta_0 \in \Theta_{j,x}, \quad \forall B \in \mathbb{N}.$$

Proof of Lemma 7. Let $j \in \mathbb{N}_0$,

$$(34) \quad \tilde{\Theta}_{j,x} = \left\{ \theta \in \mathbb{R}^d : \exists \theta' \in \Theta_{j+1,x} \text{ such that } \|\theta - \theta'\| \leq \sup_{\tilde{\theta} \in \Theta_{j+1,x}} \|\nabla F(\tilde{\theta})\| + 1 \right\}$$

and, for all $B \in \mathbb{N}$, let

$$\Omega_{j,B}^{(1)} = \left\{ \omega \in \Omega : \max_{k \in \{0,1\}} \sup_{\theta \in \tilde{\Theta}_{j,x}} \|\nabla^k F_{n_B}^\omega(\theta) - \nabla^k F(\theta)\| \leq 1/2 \right\}.$$

Under A1 the function F is strictly convex on $\mathbb{R}^d$ and thus the set $\Theta_{j+1,x}$ is convex and bounded. Therefore, since by Lemma 2 the function F is also continuous on $\mathbb{R}^d$, the set $\Theta_{j+1,x}$ is compact and such that $\theta_\star \in \dot{\Theta}_{j+1,x}$.

To show the second part of lemma remark that since $\Theta_{j+1,x}$ is a compact (and convex) set and F is continuous by Lemma 2, it follows that we have

$$\sup_{\tilde{\theta} \in \Theta_{j+1,x}} \|\nabla F(\tilde{\theta})\| < \infty$$

and thus $\tilde{\Theta}_{j,x}$ is also a compact (and convex) set. Consequently, by Lemma 3, we have $\lim_{B \rightarrow \infty} \mathbb{P}(\Omega_{j,B}^{(1)}) = 1$.

To proceed further we let $B \in \mathbb{N}$, $\omega \in \Omega_{j,B}^{(1)}$ and $\theta_0 \in \Theta_{j,x}$, and we show that $\theta_{\theta_0,B,t}^\omega \in \Theta_{j+1,x}$ for all $t \in \{0, \dots, T_{\theta_0,B}^\omega\}$. We assume below that $T_{\theta_0,B}^\omega > 0$ since otherwise the result is trivial (since by $\theta_0 \in \Theta_{j,x} \subset \Theta_{j+1,x}$).

To do so let $\theta \in \tilde{\Theta}_{j,x}$ be such that $F_{n_B}^\omega(\theta) \leq F_{n_B}^\omega(\theta_0)$ and note that

$$F(\theta) - 1/2 \leq F_{n_B}^\omega(\theta) \leq F_{n_B}^\omega(\theta_0) \leq F(\theta_0) + 1/2 \leq F(x) + j + 1/2$$

showing that $F(\theta) \leq F(x) + j + 1$ and thus that

$$(35) \quad \{\theta \in \tilde{\Theta}_{j,x} : F_{n_B}^\omega(\theta) \leq F_{n_B}^\omega(\theta_0)\} \subset \Theta_{j+1,x}.$$

We now let $t \in \{0, \dots, T_{\theta_0,B}^\omega - 1\}$ and remark that Algorithm 3 ensures that we have

$$\|\theta_{\theta_0,B,t+1}^\omega - \theta_{\theta_0,B,t}^\omega\| \leq \|\nabla F_{n_B}^\omega(\theta_{\theta_0,B,t}^\omega)\|.$$

Hence, if $\theta_{\theta_0,B,t}^\omega \in \Theta_{j+1,x} \subset \tilde{\Theta}_{j,x}$, we have

$$\|\theta_{\theta_0,B,t+1}^\omega - \theta_{\theta_0,B,t}^\omega\| \leq \|\nabla F(\theta_{\theta_0,B,t}^\omega)\| + 1/2 \leq \sup_{\tilde{\theta} \in \Theta_{j+1,x}} \|\nabla F(\tilde{\theta})\| + 1$$

showing that $\theta_{\theta_0,B,t+1}^\omega \in \tilde{\Theta}_{j,x}$. Therefore, since Algorithm 3 ensures that we have

$$F_{n_B}^\omega(\theta_{\theta_0,B,t+1}^\omega) \leq F_{n_B}^\omega(\theta_{\theta_0,B,t}^\omega) \leq F_{n_B}^\omega(\theta_0),$$

it follows from (35) that $\theta_{\theta_0,B,t+1}^\omega \in \Theta_{j+1,x}$ if $\theta_{\theta_0,B,t}^\omega \in \Theta_{j+1,x}$. Consequently, since $\theta_0 \in \Theta_{j,x} \subset \Theta_{j+1,x}$, it follows that $\theta_{\theta_0,B,t}^\omega \in \Theta_{j+1,x}$ for all $t \in \{0, \dots, T_{\theta_0,B}^\omega\}$ and the proof of the lemma is complete. $\square$

A.3.3. A second key property of Algorithm 3.

Lemma 8. Consider the set-up of Lemma 7. For all $\theta_0 \in \mathbb{R}^d$, $B \in \mathbb{N}$ and $t \in \{1, \dots, T_{\theta_0,B}\}$, let $C_{\theta_0,B,t}$ denote the computational cost needed for computing $\theta_{\theta_0,B,t}^\omega$ from $\theta_{\theta_0,B,t-1}^\omega$, with the convention that $C_{\theta_0,B,t} = 0$ when $T_{\theta_0,B}^\omega = 0$. Then, for all $j \in \mathbb{N}_0$, there exist (i) a compact and convex set $K_{j+1,x} \subset \mathbb{R}^d$ such that $\Theta_{j+1,x} \subseteq K_{j+1,x}$, (ii) a constant $D_j \in (0, \infty)$ and (iii) a sequence $(\Omega_{j,B}^{(2)})_{B \geq 1}$ in $\mathcal{F}$ such that $\lim_{B \rightarrow \infty} \mathbb{P}(\Omega_{j,B}^{(2)}) = 1$, such that we have, for all $B \in \mathbb{N}$, all $\theta_0 \in \Theta_{j,x}$ and all $\omega \in \Omega_{j,B}^{(2)}$,

$$F_{n_B}^\omega(\hat{\theta}_{\theta_0,B}^\omega) - F_{n_B}^\omega(\hat{\theta}_{K_{j+1,x},n_B}^\omega) \leq \frac{D_j \|\theta_0 - \hat{\theta}_{K_{j+1,x},n_B}^\omega\|^2}{T_{\theta_0,B}^\omega}, \quad C_{\theta_0,B,t}^\omega \leq D_j n_B.$$

Proof of Lemma 8. Let $j \in \mathbb{N}_0$ and $\tilde{\Theta}_{j,x}$ be as defined in (34). Recall from the proof of Lemma 7 that $\tilde{\Theta}_{j,x}$ is a compact and convex set, and for all $B \in \mathbb{N}$ let

$$\Omega_{j,B}^{(2)} = \Omega_{j,B}^{(1)} \cap \left\{ \omega \in \Omega : F_{n_B}^\omega \text{ is strictly convex on } \tilde{\Theta}_{j,x}, \right. \\ \left. \|\nabla F_{n_B}^\omega(\theta) - \nabla F_{n_B}^\omega(\theta')\| \leq 2\mathbb{E}[M'_{\tilde{\Theta}_{j,x}}(Z)]\|\theta - \theta'\| \text{ for all } (\theta, \theta') \in \tilde{\Theta}_{j,x}^2 \right\}$$

with $\Omega_{j,B}^{(1)}$ as in the statement of Lemma 7 and with the function $M'_{\tilde{\Theta}_{j,x}}(\cdot)$ as in Lemma 1 (when $\Theta = \tilde{\Theta}_{j,x}$). Remark that, by Lemma 1, Corollary 1 and Lemma 7, we have $\lim_{B \rightarrow \infty} \mathbb{P}(\Omega_{j,B}^{(2)}) = 1$.

To prove the result of the lemma let $L_j = (2\mathbb{E}[M'_{\tilde{\Theta}_{j,x}}(Z)] \vee 1)$, $\theta_0 \in \Theta_{j,x}$, $B \in \mathbb{N}$ and $\omega \in \Omega_{j,B}^{(2)}$. Remark that the function $F_{n_B}^\omega$ is strictly convex and L_j -smooth on

the compact and convex set $\tilde{\Theta}_{j,x}$, and that since $\omega \in \Omega_{j,B}^{(1)}$ we have $\theta_{\theta_0,B,t}^\omega \in \Theta_{j+1,x}$ for all $t \in \{0, \dots, T_{\theta_0,B}^\omega\}$.

Assume first that $T_{\theta_0,B}^\omega > 0$ and for all $t \in \{1, \dots, T_{\theta_0,B}^\omega\}$ let v_t^ω be such that $\theta_{\theta_0,B,t}^\omega = \theta_{\theta_0,B,t-1}^\omega + v_t^\omega \nabla F_{n_B}(\theta_{\theta_0,B,t-1}^\omega)$. Then, by applying Proposition 6 with $g = F_{n_B}$, $K = \tilde{\Theta}_{j,x}$, $\tilde{K} = \Theta_{j+1,x}$ and $L_K = L_j$, it follows that $v_t^\omega \geq \beta/L_j$ for all $t \in \{1, \dots, T_{\theta_0,B}^\omega\}$. From this result, we readily obtain that

$$C_{\theta_0,B,t}^\omega \leq D_j n_B, \quad D_j = (\beta^{k_j} + 1)(C_{\text{eval}} + C_{\text{grad}}), \quad k_j = \inf\{k \in \mathbb{N} : \beta^k < \beta/L_j\}.$$

In addition, by Proposition 5, we have

$$F_{n_B}^\omega(\hat{\theta}_{\theta_0,B}^\omega) - F_{n_B}^\omega(\hat{\theta}_{\tilde{\Theta}_{j,x},n_B}^\omega) \leq \frac{\beta \|\theta_0 - \hat{\theta}_{\tilde{\Theta}_{j,x},n_B}^\omega\|^2}{2L_j T_{\theta_0,B}^\omega}.$$

Noting that this latter inequality also holds if $T_{\theta_0,B}^\omega = 0$, the proof of the lemma is complete. $\square$

A.3.4. A third key property of Algorithm 3.

Lemma 9. *Consider the set-up of Lemma 8. Assume in addition that there exists a constant $c \in (0, 1)$ such that $\mathbb{P}(\tilde{B}_B \geq cB) = 1$ for all $B \in \mathbb{N}$ and that $\lim_{B \rightarrow \infty} n_B/B = 0$ and that $\lim_{B \rightarrow \infty} \tau_B = 0$. Then, for all $j \in \mathbb{N}_0$, there exist a constant $D'_j \in (0, \infty)$ and a sequence $(\Omega_{j,B}^{(3)})_{B \geq 1}$ in $\mathcal{F}$ such that $\lim_{B \rightarrow \infty} \mathbb{P}(\Omega_{j,B}^{(3)}) = 1$ and such that, with $K_{j+1,x}$ as in Lemma 8,*

$$\|\hat{\theta}_{\theta_0,B}^\omega - \hat{\theta}_{K_{j+1,x},n_B}^\omega\| \leq \frac{D'_j \|\theta_0 - \hat{\theta}_{K_{j+1,x},n_B}^\omega\|}{\sqrt{T_{\theta_0,B}^\omega}}, \quad \forall \omega \in \Omega_{j,B}^{(3)}, \quad \forall \theta_0 \in \Theta_{j,x}, \quad \forall B \in \mathbb{N}.$$

Proof of Lemma 9. Let $j \in \mathbb{N}_0$, and to simplify the notation in what follows let $K_x = K_{j+1,x}$ and, for all $\theta \in \mathbb{R}^d$ and $n \geq 1$, let $H_n(\theta) = \nabla^2 F_n(\theta)$.

We start by showing that

$$(36) \quad \sup_{\theta_0 \in \Theta_{j,x}} \|\hat{\theta}_{\theta_0,B} - \hat{\theta}_{K_x,n_B}\| = o_{\mathbb{P}}(1).$$

To this aim let $M'_{K_x}(\cdot)$ be as defined in Lemma 1 when $\Theta = K_x$,

$$\lambda_{K_x} = \frac{1}{2} \inf_{\theta \in K_x} \lambda_{\min}(\mathbb{E}[\nabla^2 f(\theta, Z)])$$

and, for all $\epsilon > 0$ and $B \in \mathbb{N}$, let

$$\begin{aligned} \Omega_{\epsilon,B} &= \Omega_{j,B}^{(1)} \cap \Omega_{j,B}^{(2)} \\ &\cap \left\{ \omega \in \Omega : \tilde{B}_B^\omega \geq cB, \inf_{\theta \in K_x} \lambda_{\min}(H_{n_B}^\omega(\theta)) \geq \lambda_{K_x}, \|\hat{\theta}_{K_x,n_B}^\omega - \theta_\star\| \leq \epsilon, \right. \\ &\quad \left. \hat{\theta}_{K_x,n_B}^\omega \in \overset{\circ}{K}_x, \sup_{\theta \in K_x} |F_{n_B}^\omega(\theta) - F(\theta)| \leq \epsilon, \frac{1}{n_B} \sum_{i=1}^{n_B} M'_{K_x}(Z^\omega) \leq 2\mathbb{E}[M'_{K_x}(Z)] \right\} \end{aligned}$$

with $\Omega_{j,B}^{(1)}$ as in Lemma 7 and with $\Omega_{j,B}^{(2)}$ as in Lemma 8. Remark that $\theta_\star \in \overset{\circ}{K}_x$ (see Lemmas 7–8) and thus, by Lemmas 1, 3, 4, 5, 7 and 8, we have $\liminf_{B \rightarrow \infty} \mathbb{P}(\Omega_{\epsilon,B}) = 1$ for all $\epsilon > 0$.

We now let $\theta_0 \in \Theta_{j,x}$, $\epsilon > 0$, $B \in \mathbb{N}$ and $\omega \in \Omega_{\epsilon,B}$, and assume first that $\|\nabla F_{n_B}^\omega(\hat{\theta}_{\theta_0,B}^\omega)\| \leq \tau_B$. By Lemma 2 and under A1 we have $\lambda_{K_x} > 0$, implying that

the function $F_{n_B}^\omega$ is strictly convex on the compact and convex set K_x . Consequently, since $\hat{\theta}_{K_x, n_B}^\omega \in \overset{\circ}{K}_x$, we have $\nabla F_{n_B}^\omega(\hat{\theta}_{K_x, n_B}^\omega) = 0$. Therefore, using Taylor's theorem, the fact that $\|Ax\| \geq \lambda_{\min}(A)\|x\|$ for any $d \times d$ symmetric matrix A and $x \in \mathbb{R}^d$, and noting that $\hat{\theta}_{\theta_0, B}^\omega \in \Theta_{j+1, x} \subset K_x$, we have

$$\tau_B \geq \|\nabla F_{n_B}^\omega(\hat{\theta}_{\theta_0, B}^\omega)\| \geq \lambda_{K_x} \|\hat{\theta}_{\theta_0, B}^\omega - \hat{\theta}_{K_x, n_B}^\omega\|$$

showing that

$$(37) \quad \|\hat{\theta}_{\theta_0, B}^\omega - \hat{\theta}_{K_x, n_B}^\omega\| \leq \frac{\tau_B}{\lambda_{K_x}}, \quad \text{if } \|\nabla F_{n_B}^\omega(\hat{\theta}_{\theta_0, B}^\omega)\| \leq \tau_B.$$

Assume now that $\|\nabla F_{n_B}^\omega(\hat{\theta}_{\theta_0, B}^\omega)\| > \tau_B$, in which case we have

$$(38) \quad T_{\theta_0, B}^\omega \geq \frac{cB}{D_j n_B}$$

with $D_j < \infty$ as in Lemma 8. Under A1 there exists a constant $\mu_x > 0$ such that

$$|F(\theta) - F(\theta_*)| \geq \frac{\mu_x}{2} \|\theta - \theta_*\|^2, \quad \forall \theta \in K_x$$

and thus

$$(39) \quad \begin{aligned} \frac{\mu_x}{2} \|\hat{\theta}_{\theta_0, B}^\omega - \theta_*\|^2 &\leq |F(\hat{\theta}_{\theta_0, B}^\omega) - F(\theta_*)| \\ &\leq |F(\hat{\theta}_{\theta_0, B}^\omega) - F_{n_B}^\omega(\hat{\theta}_{\theta_0, B}^\omega)| + |F_{n_B}^\omega(\hat{\theta}_{\theta_0, B}^\omega) - F(\theta_*)| \\ &\leq \epsilon + |F_{n_B}^\omega(\hat{\theta}_{\theta_0, B}^\omega) - F(\theta_*)|. \end{aligned}$$

Recalling that $\theta_* \in K_x$, we have

$$(40) \quad \begin{aligned} &|F_{n_B}^\omega(\hat{\theta}_{\theta_0, B}^\omega) - F(\theta_*)| \\ &\leq |F_{n_B}^\omega(\hat{\theta}_{\theta_0, B}^\omega) - F_{n_B}^\omega(\hat{\theta}_{K_x, n_B}^\omega)| + |F_{n_B}^\omega(\hat{\theta}_{K_x, n_B}^\omega) - F_{n_B}^\omega(\theta_*)| \\ &\quad + |F_{n_B}^\omega(\theta_*) - F(\theta_*)| \\ &\leq |F_{n_B}^\omega(\hat{\theta}_{\theta_0, B}^\omega) - F_{n_B}^\omega(\hat{\theta}_{K_x, n_B}^\omega)| \\ &\quad + \left(\frac{1}{n_B} \sum_{i=1}^{n_B} M'_{K_x}(Z_i^\omega) \right) \|\hat{\theta}_{K_x, n_B}^\omega - \theta_*\| + \sup_{\theta \in K_x} |F_{n_B}^\omega(\theta) - F(\theta)| \\ &\leq |F_{n_B}^\omega(\hat{\theta}_{\theta_0, B}^\omega) - F_{n_B}^\omega(\hat{\theta}_{K_x, n_B}^\omega)| + \epsilon(2\mathbb{E}[M'_{K_x}(Z)] + 1) \\ &\leq \frac{D_j \sup_{(\theta, \theta') \in K_x^2} \|\theta - \theta'\|^2}{T_{\theta_0, B}^\omega} + \epsilon(2\mathbb{E}[M'_{K_x}(Z)] + 1) \\ &\leq \frac{n_B D_j^2 \sup_{(\theta, \theta') \in K_x^2} \|\theta - \theta'\|^2}{cB} + \epsilon(2\mathbb{E}[M'_{K_x}(Z)] + 1), \end{aligned}$$

where the last inequality holds by (38).

By combining (39) and (40) we obtain that if $\|F_{n_B}^\omega(\hat{\theta}_{\theta_0, B}^\omega)\| > \tau_B$ then

$$\begin{aligned} &\|\hat{\theta}_{\theta_0, B}^\omega - \hat{\theta}_{K_x, n_B}^\omega\| \\ &\leq \epsilon + \sqrt{\frac{2}{\mu_x}} \left(\epsilon + \frac{n_B D_j^2 \sup_{(\theta, \theta') \in K_x^2} \|\theta - \theta'\|^2}{cB} + \epsilon(2\mathbb{E}[M'_{K_x}(Z)] + 1) \right)^{\frac{1}{2}}. \end{aligned}$$

Together with (37), this shows that for all $\epsilon > 0$, $B \in \mathbb{N}$ and $\omega \in \Omega_{\epsilon, B}$, we have

$$\sup_{\theta_0 \in \Theta_{j,x}} \|\hat{\theta}_{\theta_0, B}^\omega - \hat{\theta}_{K_x, n_B}^\omega\| \leq \epsilon + \frac{\tau_B}{\lambda_{K_x}} + \sqrt{\frac{2}{\mu_x}} \left(\epsilon + \frac{n_B D_j^2 \sup_{(\theta, \theta') \in K_x^2} \|\theta - \theta'\|^2}{cB} + \epsilon (2\mathbb{E}[M'_{K_x}(Z)] + 1) \right)^{\frac{1}{2}}$$

and (36) follows since, by assumption, $\lim_{B \rightarrow \infty} (n_B/B) = \lim_{B \rightarrow \infty} \tau_B = 0$ while $\epsilon > 0$ is arbitrary.

To proceed further, let

$$\delta = \frac{\lambda_{K_x}}{4\mathbb{E}[M'_{K_x}(Z)]}$$

and, for all $B \in \mathbb{N}$, let

$$\begin{aligned} \Omega_{j,B}^{(3)} &= \Omega_{j,B}^{(1)} \cap \Omega_{j,B}^{(2)} \\ &\cap \left\{ \omega \in \Omega : \hat{\theta}_{K_x, n_B}^\omega \in \overset{\circ}{K}_x, \sup_{\theta_0 \in \Theta_{j,x}} \|\hat{\theta}_{\theta_0, B}^\omega - \hat{\theta}_{K_x, n_B}^\omega\| \leq \delta, \inf_{\theta \in K_x} \lambda_{\min}(H_{n_B}^\omega(\theta)) \geq \lambda_{K_x}, \right. \\ &\quad \left. \sup_{(\theta, \theta') \in K_x^2 : \theta \neq \theta'} \frac{\|\nabla^2 F_{n_B}^\omega(\theta) - \nabla^2 F_{n_B}^\omega(\theta')\|}{\|\theta - \theta'\|} \leq 2\mathbb{E}[M'_{K_x}(Z)] \right\}. \end{aligned}$$

Note that, by Lemmas 1, 4, 7 and 8, and using (36), it follows that we have $\liminf_{B \rightarrow \infty} \mathbb{P}(\Omega_{j,B}^{(3)}) = 1$.

We now let $B \in \mathbb{N}$, $\theta_0 \in \Theta_{j,x}$ and $\omega \in \Omega_{j,B}^{(3)}$. Then, noting that we have $\nabla F_{n_B}^\omega(\hat{\theta}_{K_x, n_B}^\omega) = 0$ since $\hat{\theta}_{K_x, n_B}^\omega \in \overset{\circ}{K}_x$ and the function $F_{n_B}^\omega$ is strictly convex on the compact and convex set K_x , it follows from Taylor's theorem (see Theorem 5.8 and Remark 5.9 in [1]) that

$$\begin{aligned} F_{n_B}^\omega(\hat{\theta}_{\theta_0, B}^\omega) - F_{n_B}^\omega(\hat{\theta}_{K_x, n_B}^\omega) &= \frac{1}{2}(\hat{\theta}_{\theta_0, B}^\omega - \hat{\theta}_{K_x, n_B}^\omega)^\top H_{n_B}^\omega(\hat{\theta}_{K_x, n_B}^\omega)(\hat{\theta}_{\theta_0, B}^\omega - \hat{\theta}_{K_x, n_B}^\omega)^\top \\ &\quad + R_{\theta_0, B}^\omega, \end{aligned}$$

where

$$|R_{\theta_0, B}^\omega| \leq \|\hat{\theta}_{\theta_0, B}^\omega - \hat{\theta}_{K_x, n_B}^\omega\|^2 \delta \mathbb{E}[M'_{K_x}(Z)].$$

Therefore, with $D_j < \infty$ as in Lemma 8, we have

$$\begin{aligned} \frac{D_j \|\theta_0 - \hat{\theta}_{K_x, n_B}^\omega\|^2}{T_{\theta_0, B}^\omega} &\geq F_{n_B}^\omega(\hat{\theta}_{\theta_0, B}^\omega) - F_{n_B}^\omega(\hat{\theta}_{K_x, n_B}^\omega) \\ &\geq \|\hat{\theta}_{\theta_0, B}^\omega - \hat{\theta}_{K_x, n_B}^\omega\|^2 \left(\frac{\lambda_{K_x}}{2} - \delta \mathbb{E}[M'_\Theta(Z)] \right) \\ &= \frac{\lambda_{K_x}}{4} \|\hat{\theta}_{\theta_0, B}^\omega - \hat{\theta}_{K_x, n_B}^\omega\|^2 \end{aligned}$$

and the result of the lemma follows from the fact that $\lambda_{K_x} > 0$ under A1 and by Lemma 2. $\square$

A.3.5. A fourth key property of Algorithm 3.

Lemma 10. *Consider the set-up of Lemma 9. Assume in addition that for some sequence $(r_B)_{B \geq 1}$ in $(0, \infty)$ we have $\lim_{B \rightarrow \infty} (n_B r_B^2)/(\tau_B^2 B) = 0$. Then, for all $j \in \mathbb{N}_0$ and $\epsilon \in (0, 1]$, there exist a constant $\tilde{C}_{j,\epsilon} \in [1, \infty)$ and a sequence $(\Omega_{j,\epsilon,B}^{(4)})_{B \geq 1}$*

in $\mathcal{F}$, such that $\liminf_{B \rightarrow \infty} \mathbb{P}(\Omega_{j,\epsilon,B}^{(4)}) \geq 1 - \epsilon$, such that, for all $B \in \mathbb{N}$, $\omega \in \Omega_{j,\epsilon,B}^{(4)}$ and $\theta_0 \in \Theta_{j,x}$ verifying $\|\theta_0 - \hat{\theta}_{K_{j+1,x},n_B}^\omega\| \leq r_B$ (with $K_{j+1,x}$ in Lemma 8) we have

$$(41) \quad \hat{\theta}_{\theta_0,B}^\omega \in \Theta_{j+1,x},$$

$$(42) \quad \tilde{B}_{\theta_0}^\omega \geq \frac{cB}{2},$$

$$(43) \quad \|\hat{\theta}_{\theta_0,B}^\omega - \hat{\theta}_{K_{j+1,x},n_B}^\omega\| \leq \tilde{C}_{j,\epsilon} \left(\tau_B + r_B \sqrt{\frac{n_B}{B}} \right),$$

$$(44) \quad \|\hat{\theta}_{\theta_0,B}^\omega - \theta_\star\| \leq \tilde{C}_{j,\epsilon,x} \left(\tau_B + n_B^{-\frac{\alpha}{1+\alpha}} + r_B \sqrt{\frac{n_B}{B}} \right),$$

where $c \in (0, 1)$ is as in Lemma 9 and where $\alpha \in (0, 1]$ is as in A2.

Proof of Lemma 10. We let $j \in \mathbb{N}_0$ and to simplify the notation in what follows we let $K_x = K_{j+1,x}$ and, for all $\theta \in \mathbb{R}^d$ and $n \geq 1$, we let $H_n(\theta) = \nabla^2 F_n(\theta)$.

To show the result of the lemma, for all $B \in \mathbb{N}$ and $\theta_0 \in \mathbb{R}^d$ we let

$$(\tilde{\theta}_{\theta_0,B}, B'_{\theta_0}) = \text{Algorithm 3}(\theta_0, n_B, \tilde{B}_B, 0, \beta)$$

and we denote by $\{\theta'_{\theta_0,B,t}\}_{t=0}^{\tilde{T}_{\theta_0,B}}$ the sequence in $\mathbb{R}^d$ generated by Algorithm 3 to compute $\tilde{\theta}_{\theta_0,B}$ from θ_0 (which is therefore such that $\theta'_{\theta_0,B,0} = \theta_0$ and such that $\theta'_{\theta_0,B,\tilde{T}_{\theta_0,B}} = \tilde{\theta}_{\theta_0,B}$). In addition, we let

$$\lambda_{K_x} = \frac{1}{2} \inf_{\theta \in K_x} \lambda_{\min}(\mathbb{E}[\nabla^2 f(\theta, Z)]), \quad \bar{\lambda}_{K_x} = 2 \sup_{\theta \in K_x} \lambda_{\max}(\mathbb{E}[\nabla^2 f(\theta, Z)])$$

and, for all $\epsilon > 0$, we let $\tilde{D}_\epsilon < \infty$ be such that

$$(45) \quad \liminf_{B \rightarrow \infty} \left(\|\hat{\theta}_{K_x,n_B}^\omega - \theta_\star\| \leq \tilde{D}_\epsilon n_B^{-\frac{\alpha}{1+\alpha}} \right) \geq 1 - \epsilon.$$

Remark that such a finite constant $\tilde{D}_\epsilon$ exists for all $\epsilon > 0$ by Lemma 5.

Finally, for all $B \in \mathbb{N}$ and $\epsilon > 0$ we let

$$\begin{aligned} \tilde{\Omega}_{j,\epsilon,B}^{(4)} &= \Omega_{j,B}^{(1)} \cap \Omega_{j,B}^{(2)} \cap \Omega_{j,B}^{(3)} \\ &\cap \left\{ \omega \in \Omega : \hat{\theta}_{K_x,n_B}^\omega \in \overset{\circ}{K}_x, \tilde{B}_B^\omega \geq cB, \inf_{\theta \in K_x} \lambda_{\min}(H_{n_B}^\omega(\theta)) \geq \lambda_{K_x}, \right. \\ &\quad \left. \|\hat{\theta}_{K_x,n_B}^\omega - \theta_\star\| \leq \tilde{D}_\epsilon n_B^{-\frac{\alpha}{1+\alpha}}, \sup_{\theta \in K_x} \lambda_{\max}(H_{n_B}^\omega(\theta)) \leq \bar{\lambda}_{K_x} \right\} \end{aligned}$$

with $c \in (0, 1)$ as in Lemma 9, $\Omega_{j,B}^{(1)}$ as in Lemma 7, $\Omega_{j,B}^{(2)}$ as in Lemma 8 and $\Omega_{j,B}^{(3)}$ as in Lemma 9. Remark that $\theta_\star \in \overset{\circ}{K}_x$ (see Lemmas 7–8) and thus, by Lemma 4, Lemmas 7–9 and by (45), we have $\liminf_{B \rightarrow \infty} \mathbb{P}(\tilde{\Omega}_{j,\epsilon,B}^{(4)}) \geq 1 - \epsilon$ for all $\epsilon > 0$. In addition, remark that there exists a constant $D'_j < \infty$ such that

$$(46) \quad \begin{aligned} \|\tilde{\theta}_{\theta_0,B}^\omega - \hat{\theta}_{K_x,n_B}^\omega\| &\leq D'_j \|\theta_0 - \hat{\theta}_{K_x,n_B}^\omega\| \sqrt{\frac{n_B}{B}}, \\ \forall \omega \in \tilde{\Omega}_{j,\epsilon,B}^{(4)}, \quad \forall \theta_0 \in \Theta_{j,x}, \quad \forall B \in \mathbb{N}, \quad \forall \epsilon > 0. \end{aligned}$$

We now let $\epsilon > 0$, $B \in \mathbb{N}$, $\omega \in \tilde{\Omega}_{j,\epsilon,B}^{(4)}$ and $\theta_0 \in \Theta_{j,x}$ be such that $\|\theta_0 - \hat{\theta}_{K_x,n_B}^\omega\| \leq r_B$. Assume first that

$$\|\nabla F_{n_B}^\omega(\hat{\theta}_{\theta_0,B}^\omega)\| \leq \tau_B$$

and remark that $\nabla F_{n_B}^\omega(\hat{\theta}_{K_x, n_B}^\omega) = 0$ since $F_{n_B}^\omega$ is strictly convex on the convex set K_x (as $\lambda_{K_x} > 0$ under A1 and by Lemma 2) and since $\hat{\theta}_{K_x, n_B}^\omega \in \overset{\circ}{K}_x$.

Then, using Taylor's theorem, the fact that $\|Ax\| \geq \lambda_{\min}(A)\|x\|$ for any $d \times d$ symmetric matrix A and $x \in \mathbb{R}^d$, we have

$$(47) \quad \tau_B \geq \|\nabla F_{n_B}^\omega(\hat{\theta}_{\theta_0, B}^\omega)\| \geq \lambda_{K_x} \|\hat{\theta}_{\theta_0, B}^\omega - \hat{\theta}_{K_x, n_B}^\omega\| \Leftrightarrow \|\hat{\theta}_{\theta_0, B}^\omega - \hat{\theta}_{K_x, n_B}^\omega\| \leq \frac{\tau_B}{\lambda_{K_x}}.$$

Assume now that $\|\nabla F_{n_B}^\omega(\hat{\theta}_{\theta_0, B}^\omega)\| > \tau_B$. In this case, $\hat{\theta}_{\theta_0, B}^\omega = \tilde{\theta}_{\theta_0, B}^\omega$ and thus

$$(48) \quad \begin{aligned} \|\hat{\theta}_{\theta_0, B}^\omega - \hat{\theta}_{K_x, n_B}^\omega\| &= \|\tilde{\theta}_{\theta_0, B}^\omega - \hat{\theta}_{K_x, n_B}^\omega\| \\ &\leq D'_j \|\theta_0 - \hat{\theta}_{K_x, N_B^\omega}^\omega\| \sqrt{\frac{n_B}{B}} \\ &\leq D'_j r_B \sqrt{\frac{n_B}{B}}, \end{aligned}$$

where the first inequality holds by (46). Then, (43) follows from (47) and (48), implying that (44) holds since

$$\|\hat{\theta}_{\theta_0, B}^\omega - \theta_\star\| \leq \|\hat{\theta}_{\theta_0, B}^\omega - \hat{\theta}_{K_x, n_B}^\omega\| + \|\theta_\star - \hat{\theta}_{K_x, n_B}^\omega\| \leq \|\hat{\theta}_{\theta_0, B}^\omega - \hat{\theta}_{K_x, n_B}^\omega\| + \bar{D}_\epsilon n_B^{-\frac{\alpha}{1+\alpha}}.$$

To show (42) let $D'_j < \infty$ be as in Lemma 9 and remark that, using Taylor's theorem,

$$\begin{aligned} \|\nabla F_{n_B}^\omega(\hat{\theta}_{\theta_0, B}^\omega)\| &\leq \bar{\lambda}_{K_x} \|\hat{\theta}_{\theta_0, B}^\omega - \hat{\theta}_{K_x, n_B}^\omega\| \leq \frac{\bar{\lambda}_{K_x} D'_j \|\theta_0 - \hat{\theta}_{K_x, n_B}^\omega\|}{\sqrt{T_{\theta_0, B}^\omega}} \\ &\leq \tau_B \bar{\lambda}_{K_x} D'_j \sqrt{\frac{r_B^2}{\tau_B^2 T_{\theta_0, B}^\omega}} \end{aligned}$$

so that

$$(49) \quad T_{\theta_0, B}^\omega \leq \min \left(\tilde{T}_{\theta_0, B}^\omega, (r_B/\tau_B)^2 (\bar{\lambda}_{K_x} D'_j)^2 + 1 \right).$$

To proceed further let $D_j < \infty$ be as in Lemma 8 and remark that

$$\tilde{T}_{\theta_0, B}^\omega \geq \frac{\tilde{B}_B^\omega}{n_B D_j} \geq \frac{cB}{n_B D_j}.$$

Since by assumption, $\lim_{B \rightarrow \infty} (r_B/\tau_B)^2 n_B/B = 0$, it follows from (49) that for B large enough we have

$$T_{\theta_0, B}^\omega \leq (r_B/\tau_B)^2 (\bar{\lambda}_{K_x} D'_j)^2$$

and thus

$$\tilde{B}_{\theta_0}^\omega \geq \tilde{B}_B^\omega - T_{\theta_0, B}^\omega n_B D_j \geq cB - \frac{n_B r_B^2}{\tau_B^2} D_j (\bar{\lambda}_{K_x} D'_j)^2 = B \left(c - \frac{n_B r_B^2}{\tau_B^2 B} D_j (\bar{\lambda}_{K_x} D'_j)^2 \right).$$

Therefore, since $\lim_{B \rightarrow \infty} n_B r_B^2 / (B \tau_B^2) = 0$ by assumption, it follows that (42) holds assuming that B is sufficiently large. The result of the lemma follows. $\square$

A.4. Proof of Theorem 1.

Proof of Theorem 1. Let $\epsilon \in (0, 1)$. Then, by Lemma 5 and by applying Lemmas 8–9 with $j = 0$, $x = \theta_0$, $c = 1$ and with

$$n_B = n(B), \quad \tau_B = 0, \quad \forall B \in \mathbb{N},$$

it directly follows that there exists a sequence $(\Omega_{\epsilon, B})_{B \geq 1}$ in $\mathcal{F}$, for which we have $\liminf_{B \rightarrow \infty} \mathbb{P}(\Omega_{\epsilon, B}) \geq 1 - \epsilon$, and a constant $C_\epsilon \in [1, \infty)$, for which we have

$$\|\tilde{\theta}_B^\omega - \theta_\star\| \leq C_\epsilon \left(n(B)^{-\frac{\alpha}{1+\alpha}} + \sqrt{\frac{n_B}{B}} \right), \quad \forall \omega \in \Omega_{\epsilon, B}, \quad \forall B \in \mathbb{N}.$$

The result of the theorem follows. $\square$

A.5. Proof of Theorem 2.

A.5.1. Proof of the first part of Theorem 2.

Proof of the first part of Theorem 2. Noting that if A2 holds for some $\alpha \in (0, 1]$ then it also holds for any $\tilde{\alpha} \in (0, \alpha]$, below we can without loss of generality assume that A2 holds for some $\alpha \leq \alpha'$.

To simplify the notation in what follows, for all $\theta \in \mathbb{R}^d$ and $n \geq 1$ we let $H_n(\theta) = \nabla^2 F_n(\theta)$ and, for all $j \in \mathbb{N}$ and $B \in \mathbb{N}$ we let $n_{j, B} = \lceil \kappa B^{\gamma_j} \rceil$ and $\tau_{j, B} = \tau B^{-\frac{\alpha'}{1+\alpha'} \gamma_j}$. In addition, for all $x \in \mathbb{R}^d$ and $j \in \mathbb{N}_0$ we let the sets $\Theta_{j, x}$ and $K_{j+1, x}$ be as defined in Lemma 7 and in Lemma 8, respectively.

Let $\epsilon \in (0, 1)$. Then, by applying Lemma 10 with $j = 1$, $x = \theta_0$, $c = 1$ and with

$$(50) \quad n_B = n_{1, B}, \quad \tau_B = \tau_{1, B}, \quad r_B = \sup_{(\theta, \theta') \in K_{2, \theta_0}} \|\theta - \theta'\|, \quad \forall B \in \mathbb{N},$$

there exist a sequence $(\Omega_{\epsilon, B})_{B \geq 1}$ in $\mathcal{F}$, such that $\liminf_{B \rightarrow \infty} \mathbb{P}(\Omega_{\epsilon, B}) \geq 1 - \epsilon$, and a constant $C_\epsilon \in [1, \infty)$ for which we have

$$(51) \quad \max \left(\|\hat{\theta}_{1, B}^\omega - \theta_\star\|, \|\hat{\theta}_{1, B}^\omega - \hat{\theta}_{K_{2, \theta_0}, n_{1, B}}^\omega\| \right) \leq C_\epsilon B^{-c_{\alpha, \delta}}, \quad \forall \omega \in \Omega_{\epsilon, B}, \quad \forall B \in \mathbb{N}$$

with

$$c_{\alpha, \delta} = \min \left(\frac{(1 - \delta)\alpha}{1 + \alpha}, \frac{\delta}{2} \right) = \frac{(1 - \delta)\alpha}{1 + \alpha}.$$

Remark that, by Lemma 6, the sequences $(n_B)_{B \geq 1}$, $(\tau_B)_{B \geq 1}$ and $(r_B)_{B \geq 1}$ defined in (50) satisfy the assumptions of Lemma 10. Then, the result of the first part of the theorem for $J = 1$ follows from (51).

We assume now that $J > 1$ and, without loss of generality, we assume that $K_{j+1, \theta_0} = K$ for all $j \in \{1, \dots, J\}$ and for some compact and convex set $K \subset \mathbb{R}^d$. In addition, we let

$$\lambda_J = \frac{1}{2} \inf_{\theta \in K} \lambda_{\min}(\mathbb{E}[\nabla^2 f(\theta, Z)])$$

and, for all $\epsilon \in (0, 1)$, we let $C'_\epsilon < \infty$ be such that

$$\liminf_{B \rightarrow \infty} \mathbb{P} \left(\|\hat{\theta}_{K, n_{j, B}} - \theta_\star\| \leq C'_\epsilon n_{j, B}^{-\frac{\alpha}{1+\alpha}} \right) \geq 1 - \frac{\epsilon}{J}, \quad \forall j \in \{1, \dots, J\}.$$

Remark that, for all $\epsilon \in (0, 1)$, such a constant $C'_\epsilon < \infty$ exists by Lemma 5. Remark also that, using Fréchet's inequality,

$$(52) \quad \liminf_{B \rightarrow \infty} \mathbb{P} \left(\max_{j \in \{1, \dots, J\}} \|\hat{\theta}_{K, n_{j, B}} - \theta_\star\| \leq C'_\epsilon n_{j, B}^{-\frac{\alpha}{1+\alpha}} \right) \geq 1 - \epsilon.$$

Then, for all $\epsilon \in (0, 1)$ and $B \in \mathbb{N}$, we let

$$\begin{aligned} \Omega'_{\epsilon, B} = & \Omega_{\epsilon, B} \cap_{j=1}^J \left(\Omega_{j, B}^{(1)} \cap \Omega_{j, B}^{(2)} \right) \\ & \cap \left\{ \omega \in \Omega : \min_{j \in \{1, \dots, J\}} \inf_{\theta \in K} \lambda_{\min}(H_{n_{j, B}}^\omega(\theta)) \geq \lambda_J, \right. \\ & \left. \max_{j \in \{1, \dots, J\}} n_{j, B}^{\frac{\alpha}{1+\alpha}} \|\hat{\theta}_{K, n_{j, B}}^\omega - \theta_\star\| \leq C'_\epsilon, \hat{\theta}_{K, n_j}^\omega \in \mathring{K} \text{ for all } j \in \{1, \dots, J\} \right\} \end{aligned}$$

with $\Omega_{\epsilon, B}$ as above, $\Omega_{j, B}^{(1)}$ as in Lemma 7 and $\Omega_{j, B}^{(2)}$ as in Lemma 8. Remark that, by Lemmas 4, 7 and 8, and by (52), we have $\liminf_{B \rightarrow \infty} \mathbb{P}(\Omega'_{\epsilon, B}) \geq 1 - \epsilon$.

Let $\epsilon \in (0, 1)$, $B \in \mathbb{N}$ and $\omega \in \Omega'_{\epsilon, B}$. Remark that $\nabla F_{n_{j, B}}^\omega(\hat{\theta}_{K, n_{j, B}}^\omega) = 0$ for all $j \in \{1, \dots, J\}$, since for all $j \in \{1, \dots, J\}$ the function $F_{n_{j, B}}^\omega$ is strictly convex on K (since $\lambda_J > 0$ under A1 and by Lemma 2) while $\hat{\theta}_{K, n_j}^\omega \in \mathring{K}$. Therefore, using Taylor's theorem, the fact that $\|Ax\| \geq \lambda_{\min}(A)\|x\|$ for any $d \times d$ symmetric matrix A and $x \in \mathbb{R}^d$, we have

$$(53) \quad F_{n_{j, B}}^\omega(\theta) - F_{n_{j, B}}^\omega(\hat{\theta}_{K, n_{j, B}}^\omega) \geq \lambda_J \|\theta - \hat{\theta}_{K, n_{j, B}}^\omega\|^2, \quad \forall \theta \in K.$$

For all $j \in \mathbb{N}$ let D_j be as in Lemma 8 and assume without loss of generality that $D_j \leq D_{j+1}$ for all $j \in \mathbb{N}$ and that $D_J/\lambda_J \geq 1$. In addition, let J_B^ω denote the smallest $j \in \{1, \dots, J\}$ such that $\hat{\theta}_{j, B}^\omega = \hat{\theta}_{J, B}^\omega$. Then, noting that $n_{j, B} \leq n_{j+1, B}$ for all $j \in \mathbb{N}$, if $J_B^\omega > 1$ we have

$$\begin{aligned} & \|\hat{\theta}_{J_B^\omega, B}^\omega - \hat{\theta}_{K, n_{J_B^\omega, B}}^\omega\|^2 \\ & \leq \frac{1}{\lambda_J} \left(F_{n_{J_B^\omega, B}}^\omega(\hat{\theta}_{J_B^\omega, B}^\omega) - F_{n_{J_B^\omega, B}}^\omega(\hat{\theta}_{K, n_{J_B^\omega, B}}^\omega) \right) \\ & \leq \frac{D_J \|\hat{\theta}_{J_B^\omega-1, B}^\omega - \hat{\theta}_{K, n_{J_B^\omega, B}}^\omega\|^2}{T_{J_B^\omega, B}^\omega \lambda_J} \\ & \leq \frac{2D_J}{\lambda_J} \left(\|\hat{\theta}_{J_B^\omega-1, B}^\omega - \hat{\theta}_{K, n_{J_B^\omega-1, B}}^\omega\|^2 + \|\hat{\theta}_{K, n_{J_B^\omega, B}}^\omega - \hat{\theta}_{K, n_{J_B^\omega-1, B}}^\omega\|^2 \right) \\ & \leq \frac{4D_J}{\lambda_J} \left(\|\hat{\theta}_{J_B^\omega-1, B}^\omega - \hat{\theta}_{K, n_{J_B^\omega-1, B}}^\omega\|^2 + (C'_\epsilon)^2 n_{1, B}^{-\frac{2\alpha}{1+2\alpha}} \right) \\ & \leq \left(\frac{4D_J}{\lambda_J} \right)^J \|\hat{\theta}_{1, B}^\omega - \hat{\theta}_{K, n_{1, B}}^\omega\|^2 + (C'_\epsilon)^2 n_{1, B}^{-\frac{2\alpha}{1+2\alpha}} \sum_{j=1}^J (4D_J/\lambda_J)^j \\ & \leq \left(\frac{4D_J}{\lambda_J} \right)^J C_\epsilon^2 B^{-2c_{\alpha, \delta}} + (C'_\epsilon)^2 n_{1, B}^{-\frac{2\alpha}{1+2\alpha}} \sum_{j=1}^J (4D_J/\lambda_J)^j, \end{aligned}$$

where the first inequality uses (53) and the last inequality uses (51).

If $J_B^\omega = 1$ then, from (51), we have

$$\|\hat{\theta}_{J_B^\omega, B}^\omega - \hat{\theta}_{K, n_{J_B^\omega, B}}^\omega\|^2 \leq C_\epsilon^2 B^{-2c_{\alpha, \delta}}.$$

Therefore,

$$\|\hat{\theta}_{J,B}^\omega - \hat{\theta}_{K,n_{J^\omega},B}^\omega\| = \|\hat{\theta}_{J_B^\omega,B}^\omega - \hat{\theta}_{K,n_{J^\omega},B}^\omega\| \leq C''_\epsilon B^{-c_{\alpha,\delta}},$$

where

$$C''_\epsilon = \left(\left(\frac{4D_J}{\lambda_J} \right)^J C_\epsilon^2 + (C'_\epsilon)^2 \sum_{j=1}^J (4D_J/\lambda_J)^j \right)^{1/2}.$$

Therefore,

$$\begin{aligned} \|\hat{\theta}_{J,B}^\omega - \theta_\star\| &\leq \|\hat{\theta}_{K,n_{J^\omega},B}^\omega - \theta_\star\| + \|\hat{\theta}_{J,B}^\omega - \hat{\theta}_{K,n_{J^\omega},B}^\omega\| \\ &\leq C'_\epsilon n_{1,B}^{-\frac{\alpha}{1+\alpha}} + C''_\epsilon B^{-c_{\alpha,\delta}} \\ &\leq (C'_\epsilon + C''_\epsilon) B^{-c_{\alpha,\delta}}. \end{aligned}$$

The result of the first part of the theorem follows. $\square$

A.5.2. Proof of the second part of Theorem 2.

Proof of the 2nd part of Theorem 2. Let $c_\alpha = \alpha/(1+\alpha)$ and, for all $j \in \mathbb{N}$ and $B \in \mathbb{N}$, let $n_{j,B} = \lceil \kappa B^{\gamma_j} \rceil$, $\tau_{j,B} = \tau B^{-c_\alpha \gamma_j}$ and $r_{j,B} = B^{-c_\alpha \gamma_j}$. Finally, let $r_{0,B} = 1$ for all $B \in \mathbb{N}$ and, for all $x \in \mathbb{R}^d$ and $j \in \mathbb{N}$, let $\Theta_{j,x}$ be as defined in Lemma 7 and $K_{j+1,x}$ be as defined in Lemma 8.

To prove the second part of the theorem we show by induction that for all $j \in \mathbb{N}$ there exist, for all $\epsilon \in (0, \infty)$, a sequence $(\Omega_{\epsilon,j,B})_{B \geq 1}$ in $\mathcal{F}$ such that $\liminf_{B \rightarrow \infty} \mathbb{P}(\Omega_{\epsilon,j,B}) \geq 1 - \epsilon$, and a constant $C_{\epsilon,j} \in [1, \infty)$ for which we have

$$(54) \quad \begin{aligned} B_j^\omega &\geq \frac{B}{C_{\epsilon,j}}, \quad \|\hat{\theta}_{j,B}^\omega - \theta_\star\| \leq C_{\epsilon,j} r_{j,B}, \\ \|\hat{\theta}_{j,B}^\omega - \hat{\theta}_{K_{j+1,\theta_0},n_{j,B}}^\omega\| &\leq C_{\epsilon,j} r_{j,B}, \quad \hat{\theta}_{j,B}^\omega \in \Theta_{j+1,\theta_0} \end{aligned}$$

for all $w \in \Omega_{\epsilon,j,B}$ and $B \in \mathbb{N}$.

To do so for all $B \in \mathbb{N}$ let $n_B = n_{1,B}$, $\tau_B = \tau_{1,B}$ and $r_B = \sup_{(\theta,\theta') \in K_{2,\theta_0}} \|\theta - \theta'\|$, and note that, by Lemma 6, these sequences $(n_B)_{B \geq 1}$, $(\tau_B)_{B \geq 1}$ and $(r_B)_{B \geq 1}$ satisfy the assumptions of Lemma 10. Therefore, by applying Lemma 10 with $c = 1$, $x = \theta_0$ and $j = 1$, it follows that for all $\epsilon \in (0, \infty)$ there exist a sequence $(\Omega_{\epsilon,1,B})_{B \geq 1}$ in $\mathcal{F}$ such that $\liminf_{B \rightarrow \infty} \mathbb{P}(\Omega_{\epsilon,1,B}) \geq 1 - \epsilon$, and a constant $C_{\epsilon,1} \in [1, \infty)$, such that (54) holds for all $w \in \Omega_{\epsilon,1,B}$ and all $B \in \mathbb{N}$ when $j = 1$.

Assume now that for some $s \in \mathbb{N}$ there exist, for all $\epsilon \in (0, \infty)$, a sequence $(\Omega_{\epsilon,s,B})_{B \geq 1}$ in $\mathcal{F}$ such that $\liminf_{B \rightarrow \infty} \mathbb{P}(\Omega_{\epsilon,s,B}) \geq 1 - \epsilon$ and a constant $C_{\epsilon,s} \in [1, \infty)$ for which (54) holds for all $w \in \Omega_{\epsilon,s,B}$ and all $B \in \mathbb{N}$ when $j = s$.

For all $\epsilon > 0$ and $B \in \mathbb{N}$ we now let $\tilde{B}_{\epsilon,s}^\omega = \max(B_s^\omega, B/C_{\epsilon,s})$ for all $\omega \in \Omega$. In addition, for all $B \in \mathbb{N}$, we let $n_B = n_{s+1,B}$, $\tau_B = \tau_{s+1,B}$ and $r_B = r_{s,B}$ and remark that, by Lemma 6, these sequences $(n_B)_{B \geq 1}$, $(\tau_B)_{B \geq 1}$ and $(r_B)_{B \geq 1}$ satisfy the assumptions of Lemma 10. Let $\epsilon > 0$. Then, by applying Lemma 10 with $c = 1/C_{\frac{\epsilon}{2},s}$, $x = \theta_0$ and $j = s+1$, it follows that there exist a sequence $(\Omega'_{\epsilon/2,s+1,B})_{B \geq 1}$ in $\mathcal{F}$ such that $\liminf_{B \rightarrow \infty} \mathbb{P}(\Omega'_{\epsilon/2,s+1,B}) \geq 1 - \epsilon/2$ and a constant

$\tilde{C}_{\epsilon/2,s+1} \in [1, \infty)$ such that, for all $B \in \mathbb{N}$ and $\omega \in \Omega_{\epsilon/2,s,B} \cap \Omega'_{\epsilon/2,s+1,B}$, we have

$$(55) \quad \begin{aligned} \hat{\theta}_{s+1,B}^\omega &\in \Theta_{s+2,\theta_0}, \quad B_{s+1}^\omega \geq \frac{B}{2C_{\frac{\epsilon}{2},s}}, \\ \|\hat{\theta}_{s+1,B}^\omega - \hat{\theta}_{K_{s+1,\theta_0},n_{s+1,B}}^\omega\| &\leq \tilde{C}_{\epsilon/2,s+1} \left(\tau_{s+1,B} + r_{s,B} \sqrt{\frac{n_{s+1,B}}{B}} \right), \\ \|\hat{\theta}_{s+1,B}^\omega - \theta_\star\| &\leq \tilde{C}_{\epsilon/2,s+1} \left(\tau_{s+1,B} + r_{s,B} \sqrt{\frac{n_{s+1,B}}{B}} \right). \end{aligned}$$

To proceed further remark that, for some constant $D < \infty$,

$$(56) \quad \tau_{s+1,B} + r_{s,B} \sqrt{\frac{n_{s+1,B}}{B}} \leq D \left(B^{-c_\alpha \gamma_{s+1}} + B^{\frac{\gamma_{s+1}-1}{2} - c_\alpha \gamma_s} \right), \quad \forall B \in \mathbb{N}.$$

To proceed further let γ'_{s+1} be as defined in Lemma 6 when $\alpha' = \alpha$, and note that γ'_{s+1} is such that

$$B^{-c_\alpha \gamma'_{s+1}} = B^{\frac{\gamma'_{s+1}-1}{2} - c_\alpha \gamma_s}, \quad \forall B \in \mathbb{N}.$$

By Lemma 6, we have $\gamma_{s+1} < \gamma'_{s+1}$ and thus, since for all $B \in \mathbb{N} \setminus \{1\}$ the mapping $\gamma \mapsto B^{-c_\alpha \gamma}$ is decreasing in γ while the mapping $\gamma \mapsto B^{\frac{\gamma-1}{2} - c_\alpha \gamma_s}$ is increasing in γ , it follows that

$$B^{\frac{\gamma_{s+1}-1}{2} - c_\alpha \gamma_s} \leq B^{-c_\alpha \gamma_{s+1}}, \quad \forall B \in \mathbb{N}$$

and thus, by (56),

$$(57) \quad \tau_{s+1,B} + r_{s,B} \sqrt{\frac{n_{s+1,B}}{B}} \leq 2DB^{-c_\alpha \gamma_{s+1}} = 2Dr_{s+1,B}, \quad \forall B \in \mathbb{N}.$$

By combining (55) and (57), it follows that there exists a constant $C_{\epsilon,s+1} \in [1, \infty]$ such that, for $j = s+1$, (54) holds for all $\omega \in \Omega_{\epsilon/2,s,B} \cap \Omega'_{\epsilon/2,s+1,B}$ and $B \in \mathbb{N}$ where, using Fréchet's inequality,

$$\begin{aligned} \liminf_{B \rightarrow \infty} \mathbb{P}(\Omega_{\epsilon/2,s,B} \cap \Omega'_{\epsilon/2,s+1,B}) &\geq \liminf_{B \rightarrow \infty} \mathbb{P}(\Omega_{\epsilon/2,s,B}) + \liminf_{B \rightarrow \infty} \mathbb{P}(\Omega'_{\epsilon/2,s+1,B}) - 1 \\ &\geq 1 - \epsilon. \end{aligned}$$

Since $\epsilon > 0$ is arbitrary, this concludes to show that for all $j \in \mathbb{N}$ there exists, for all $\epsilon \in (0, \infty)$, a sequence $(\Omega_{\epsilon,j,B})_{B \geq 1}$ in $\mathcal{F}$ such that $\liminf_{B \rightarrow \infty} \mathbb{P}(\Omega_{\epsilon,j,B}) \geq 1 - \epsilon$, and a constant $C_{\epsilon,j} \in [1, \infty)$ such that (54) holds for all $\omega \in \Omega_{\epsilon,j,B}$ and all $B \in \mathbb{N}$. The second part of the theorem follows. $\square$

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